

8-bit Arithmetic Logic Unit Design Report

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8bit ALU Design Report

◆ Outline

- Generalization
- Function
- Structure
- Assignment
- Design results
- Summary

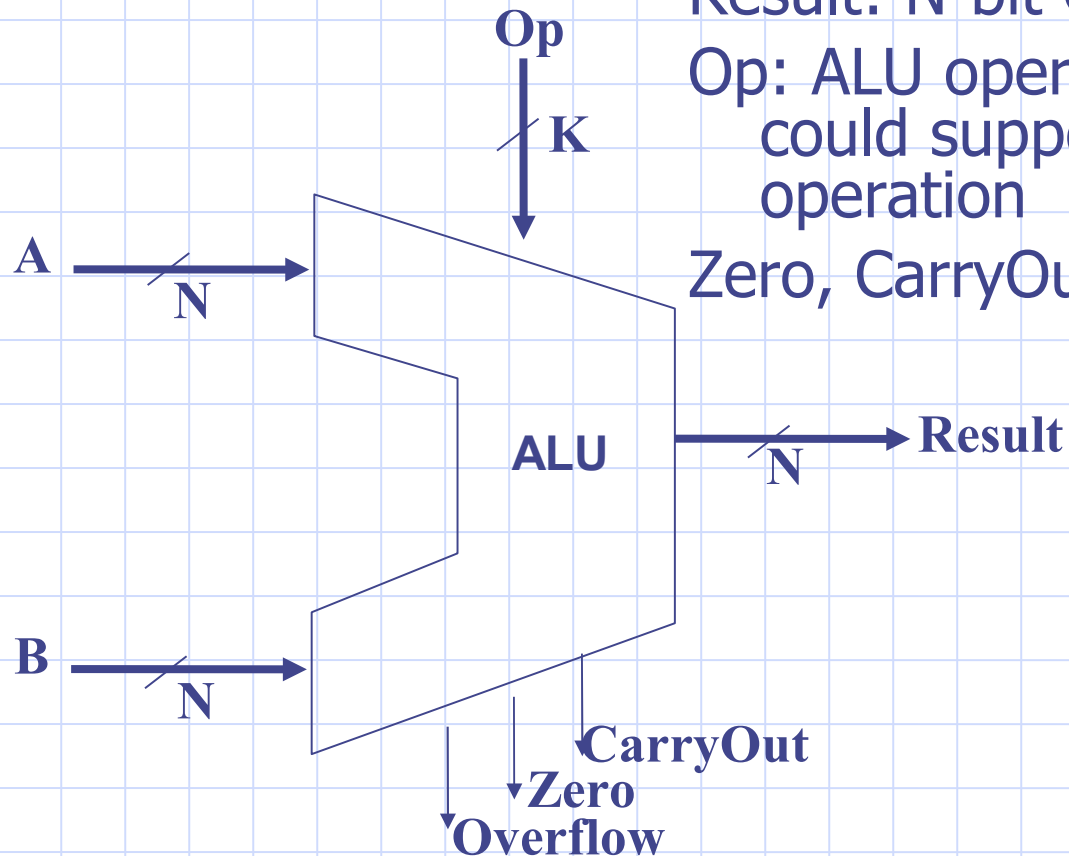
Generalization

◆ ALU (Arithmetic Logic Unit)

A critical component of the microprocessor, the core component of central processing unit. ALU comprises the combinational logic that implements logic operations such as AND and OR, and arithmetic operations such as Addition, Subtraction, and Multiplication.

Generalization

◆ ALU symbol



A and B: N-bit Inputs

Result: N-bit Output

Op: ALU operation, K-bit means
could support maximum 2^k
operation

Zero, CarryOut, Overflow: 1-bit flag

Function

◆ 8 Functions

LOGIC: And; Or; Inv

SHIFT: signed-shift left
signed-shift right

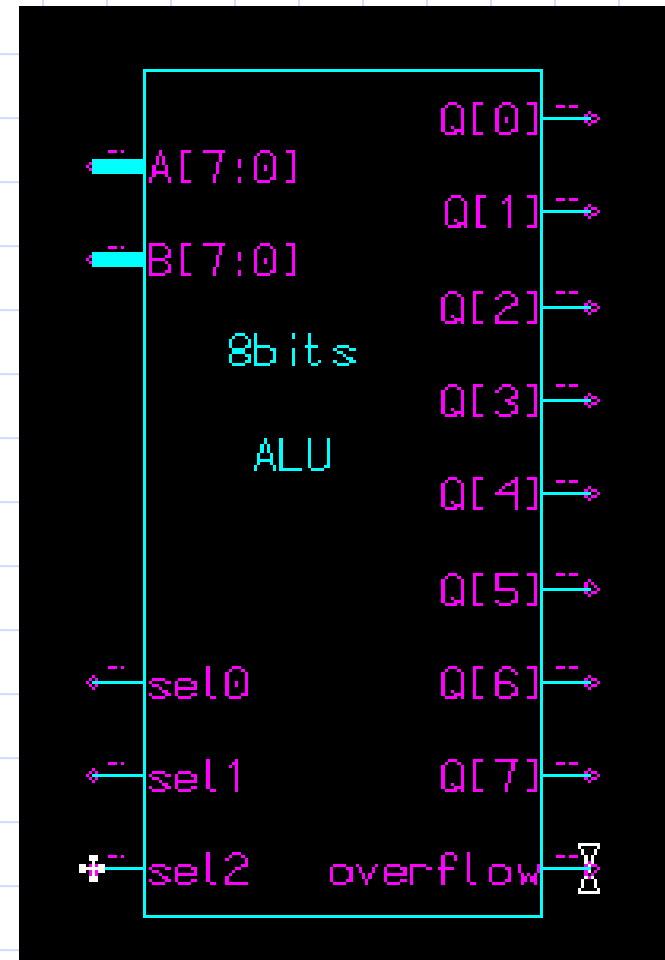
ALGORITHM: Add

Subtract

Multiply

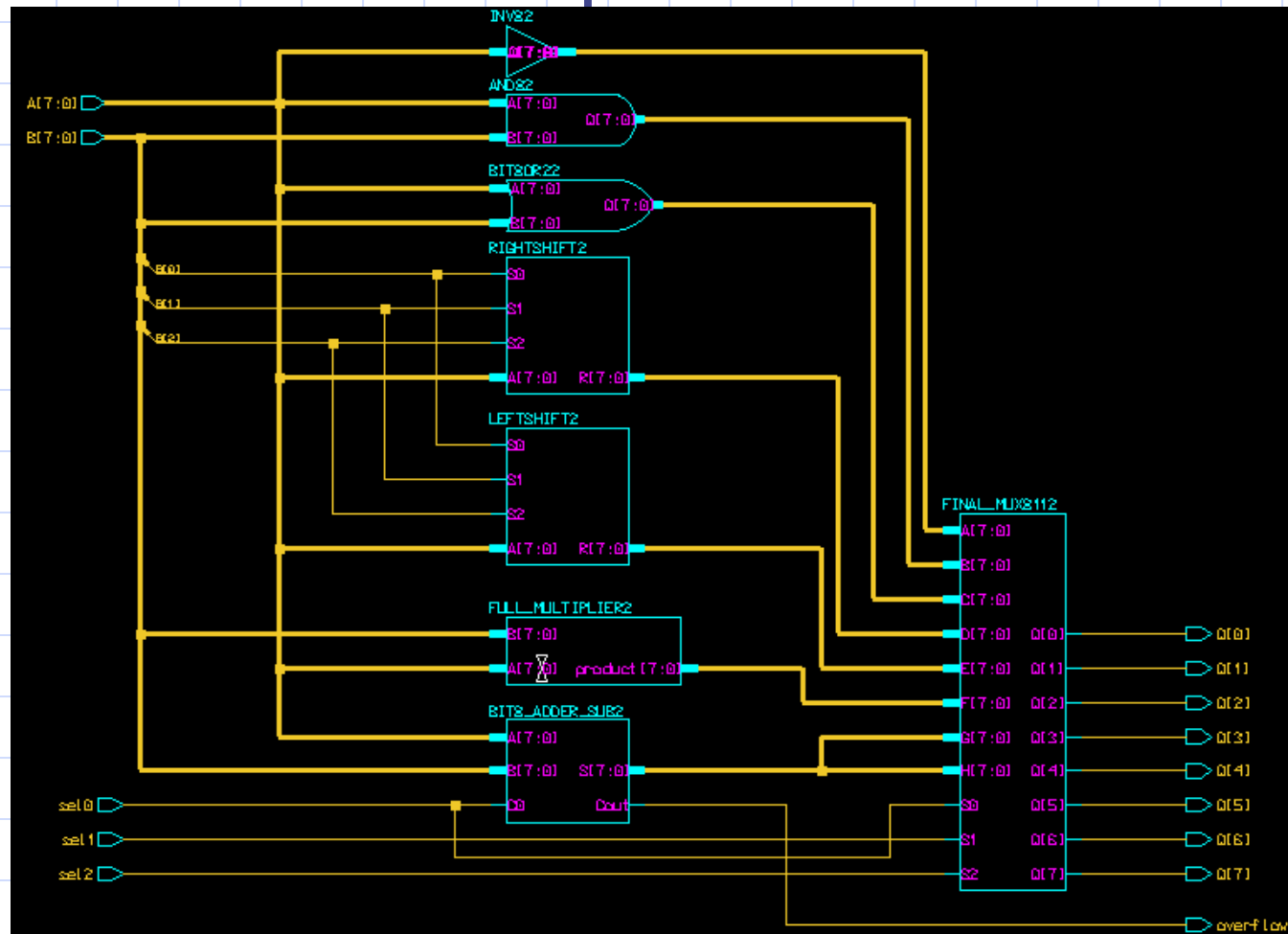
◆ 3 bit control signals

To choose the function



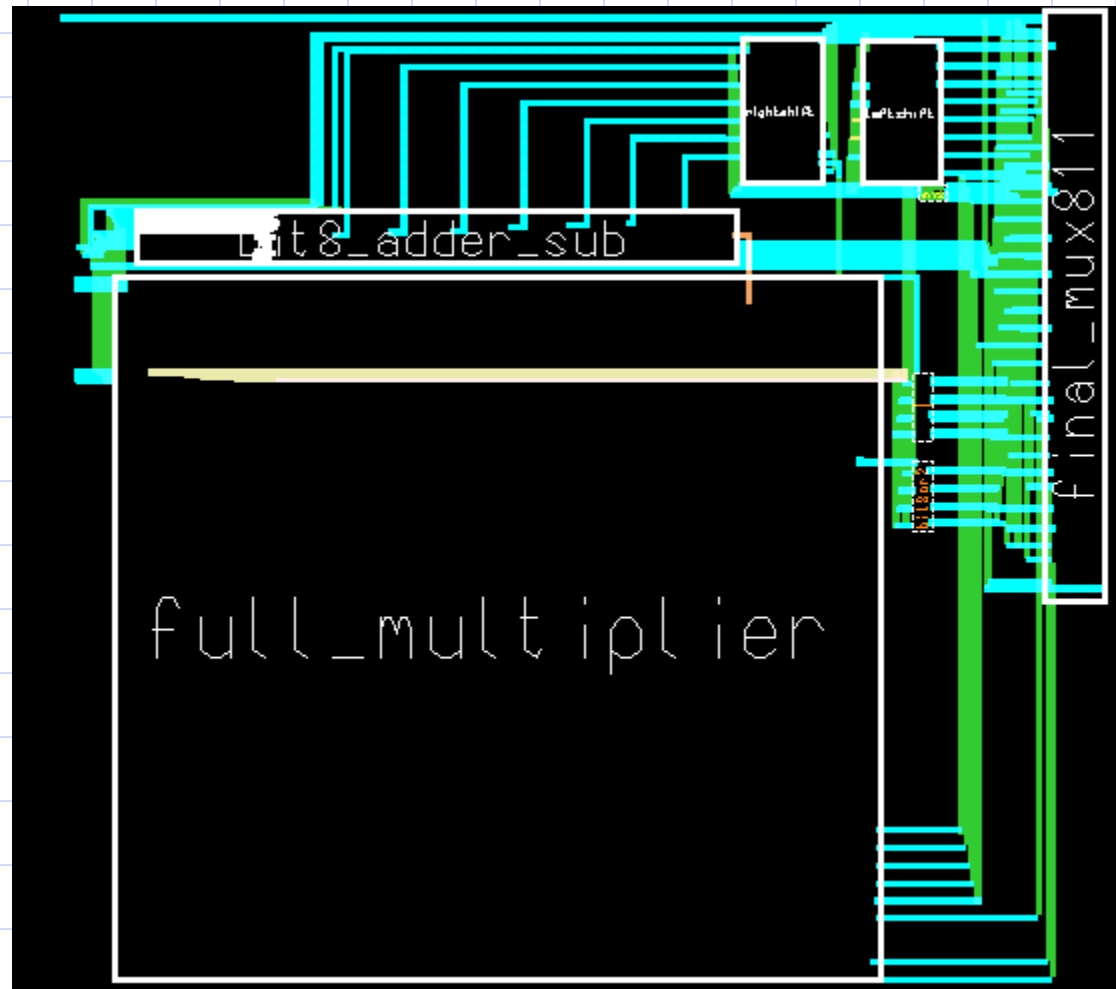
Structure

◆ Schematic – 7 components



Structure

- ◆ Layout
- place



Assignment

- ◆ Logic & Control – Wei Zhong
- ◆ Shift – Yang Zhao
- ◆ Add & Subtract – Hongxia Fang
- ◆ Multiply – Zhaobo Zhang

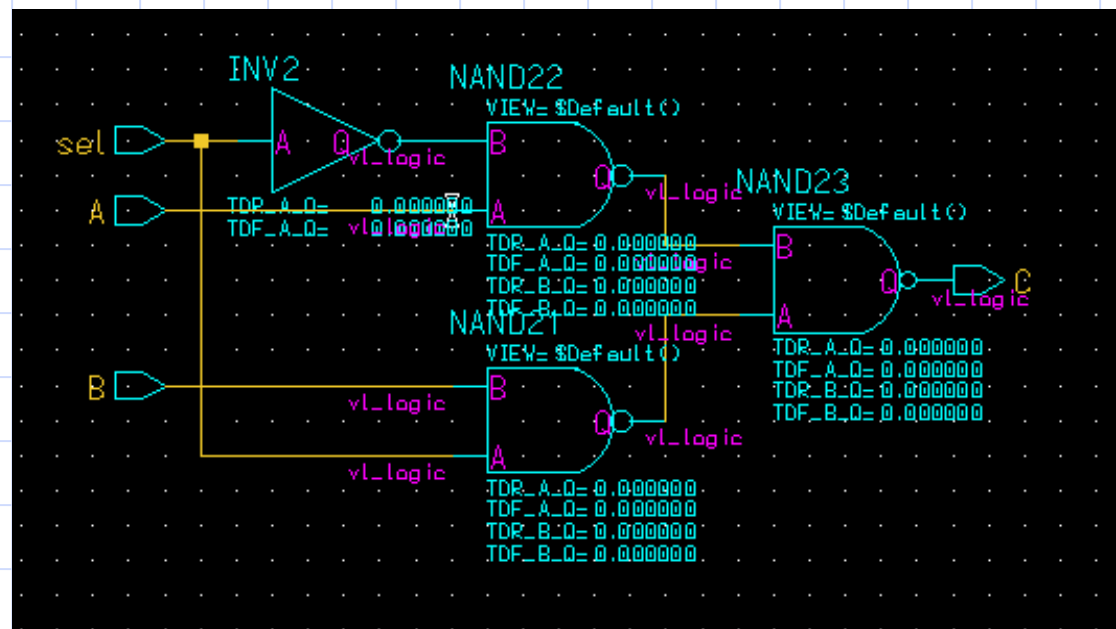
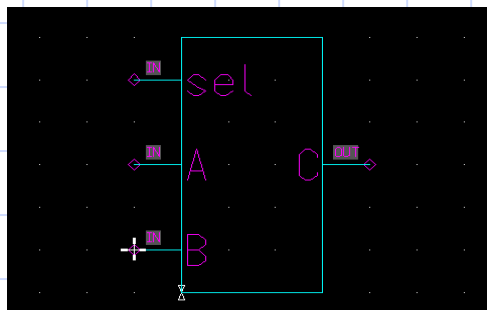
Everyone is in charge of each part's schematic simulation and layout verification

Design results

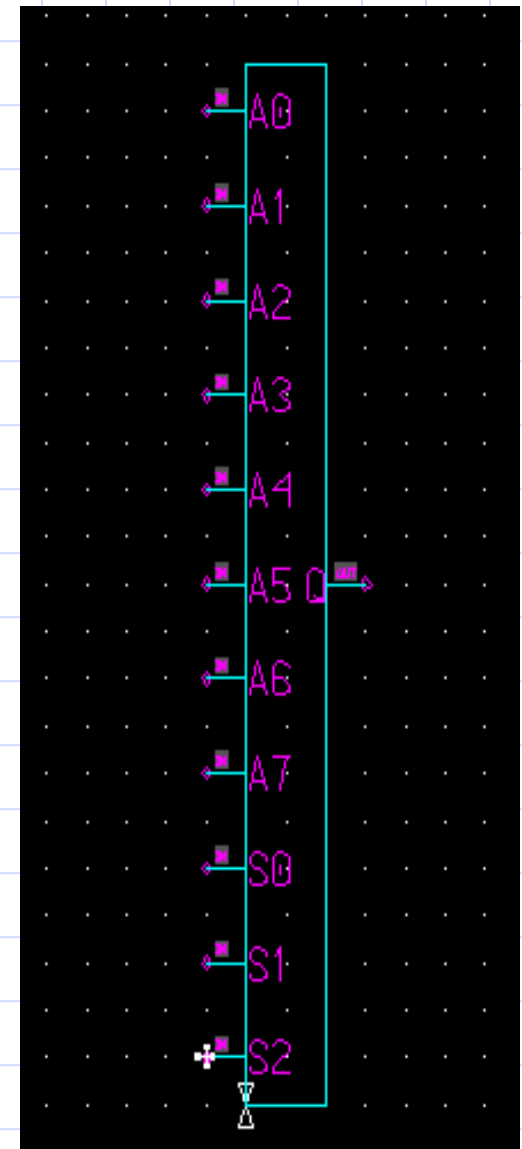
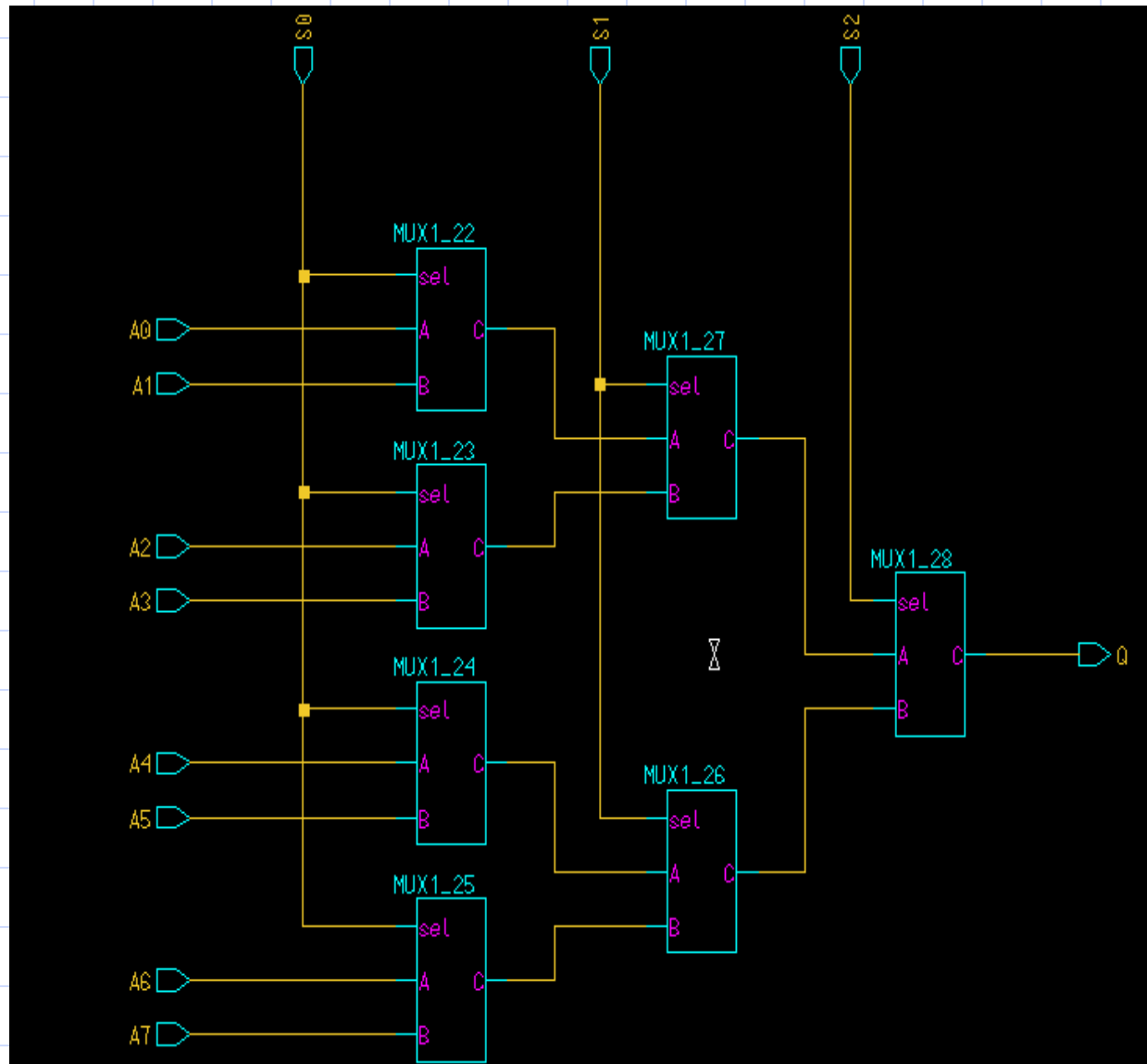
◆ 8 to 1 MUX

- Using compound gate to realize 2 to 1 MUX

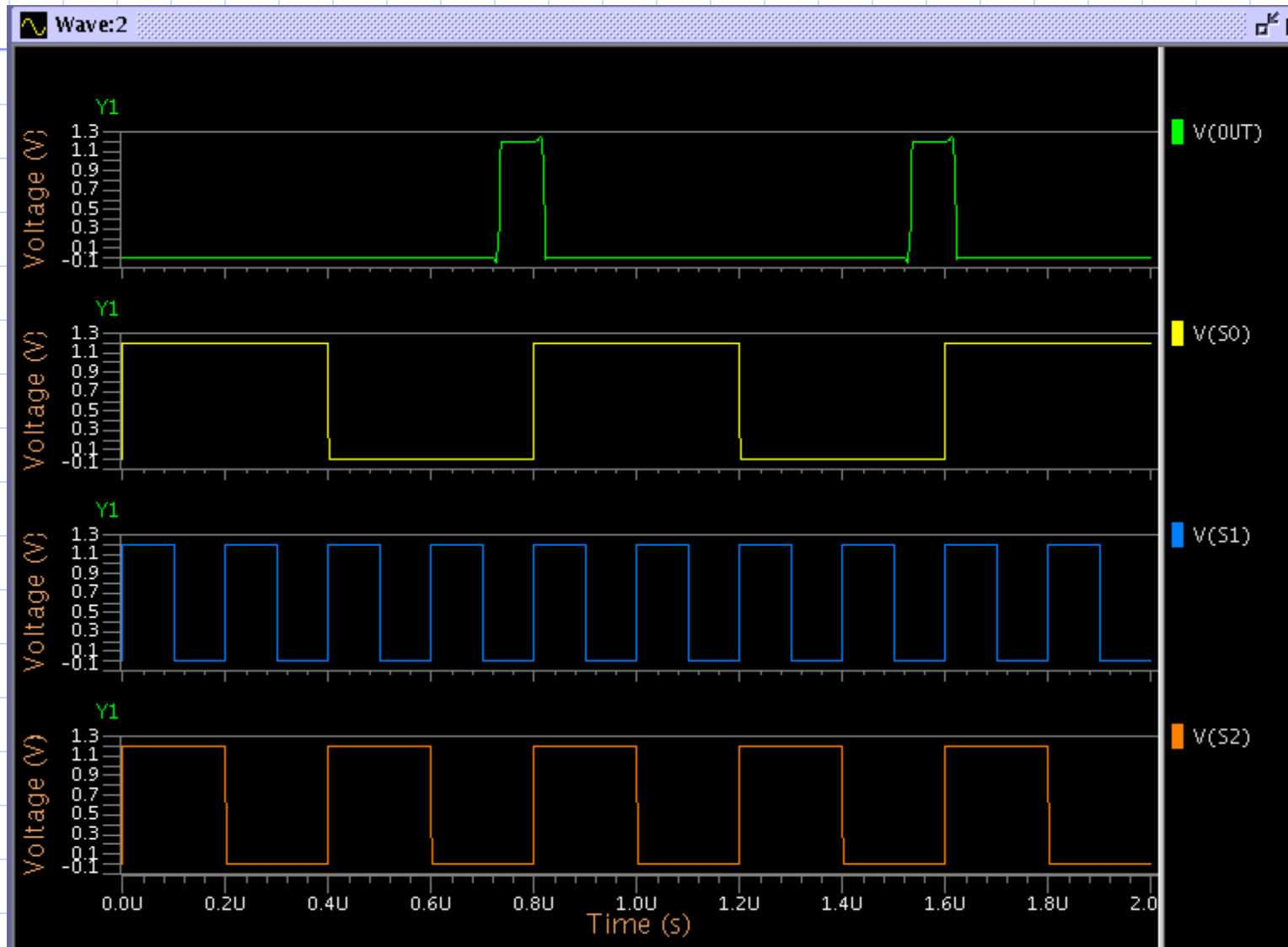
$$Y = D \cdot \bar{S} + D \cdot S$$



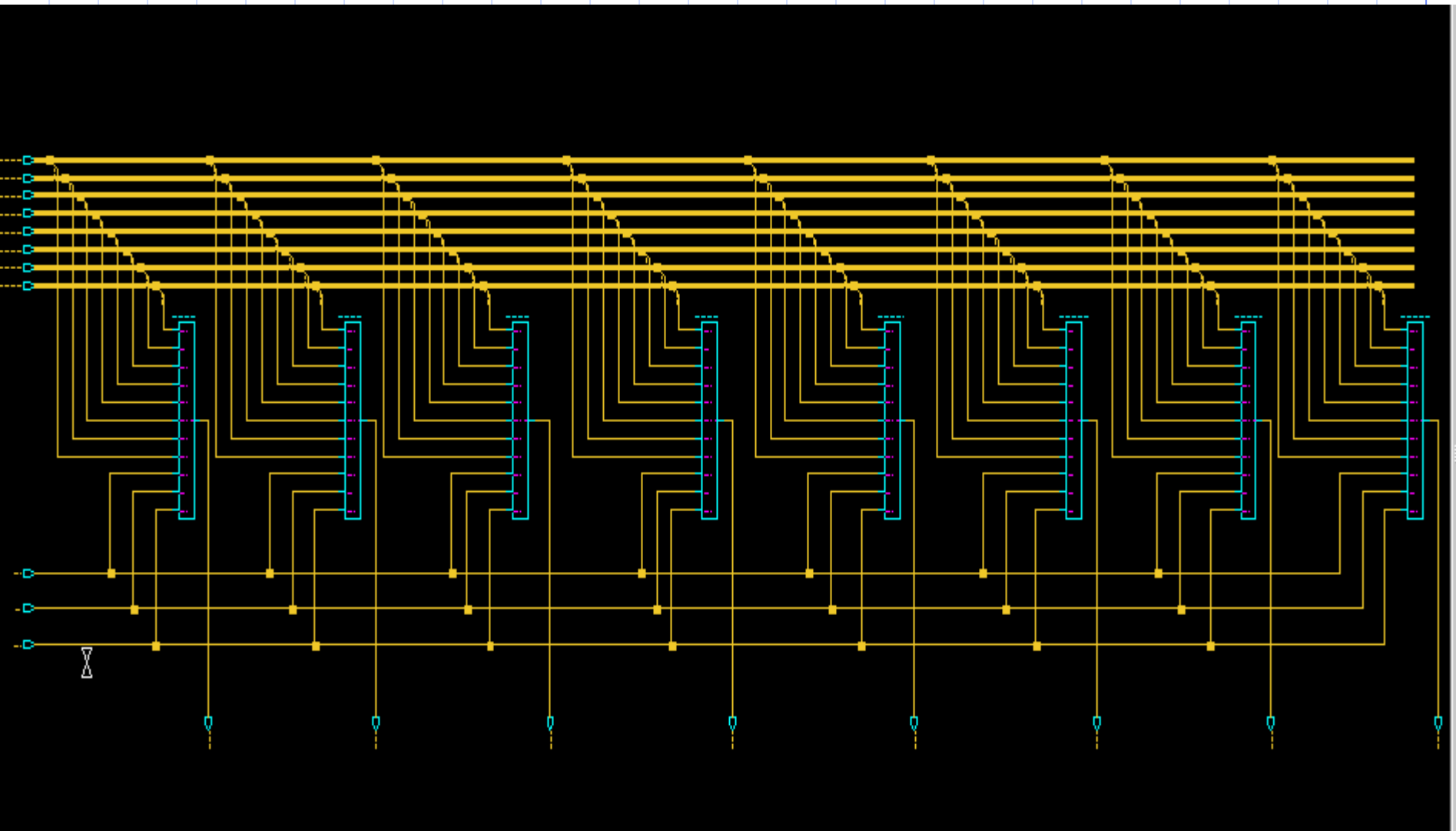
8 to 1 Mux



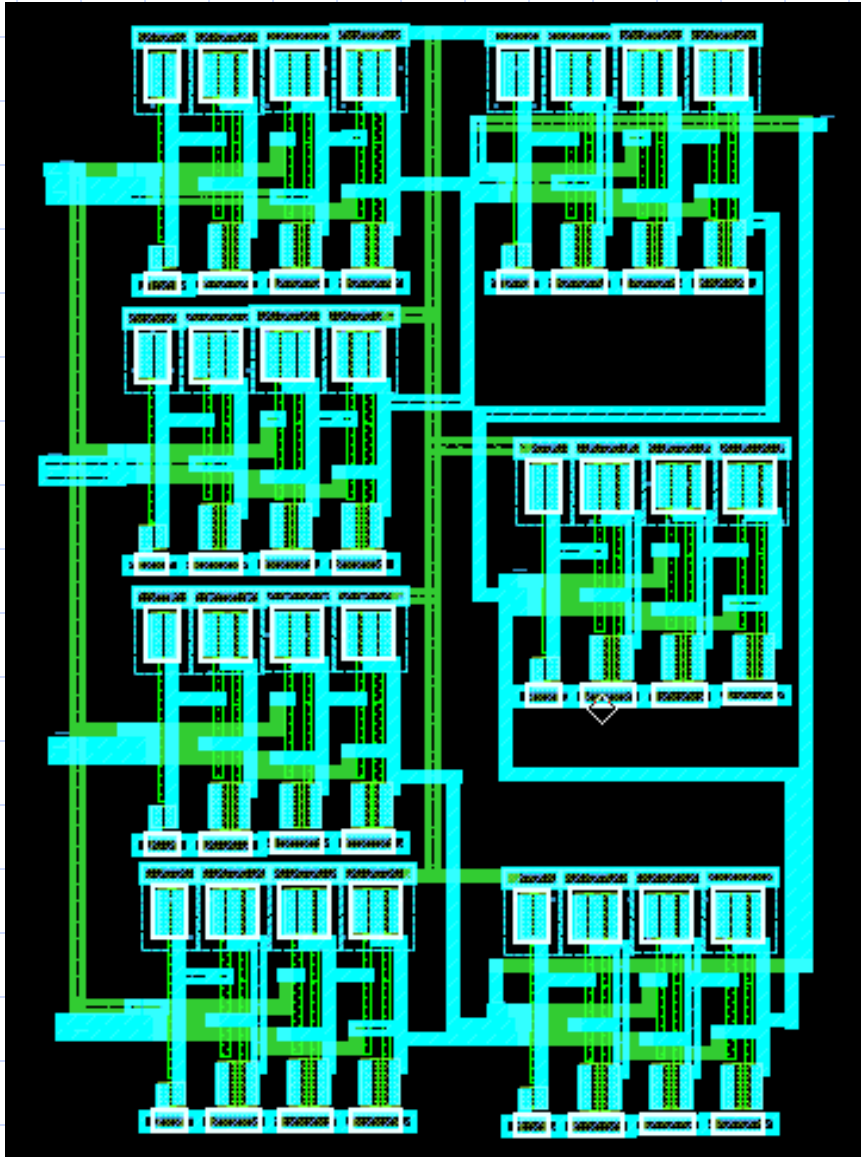
Set $A0=1$, $A1\sim A7=0$.
when $S2S1S0=000$, $Out=1$



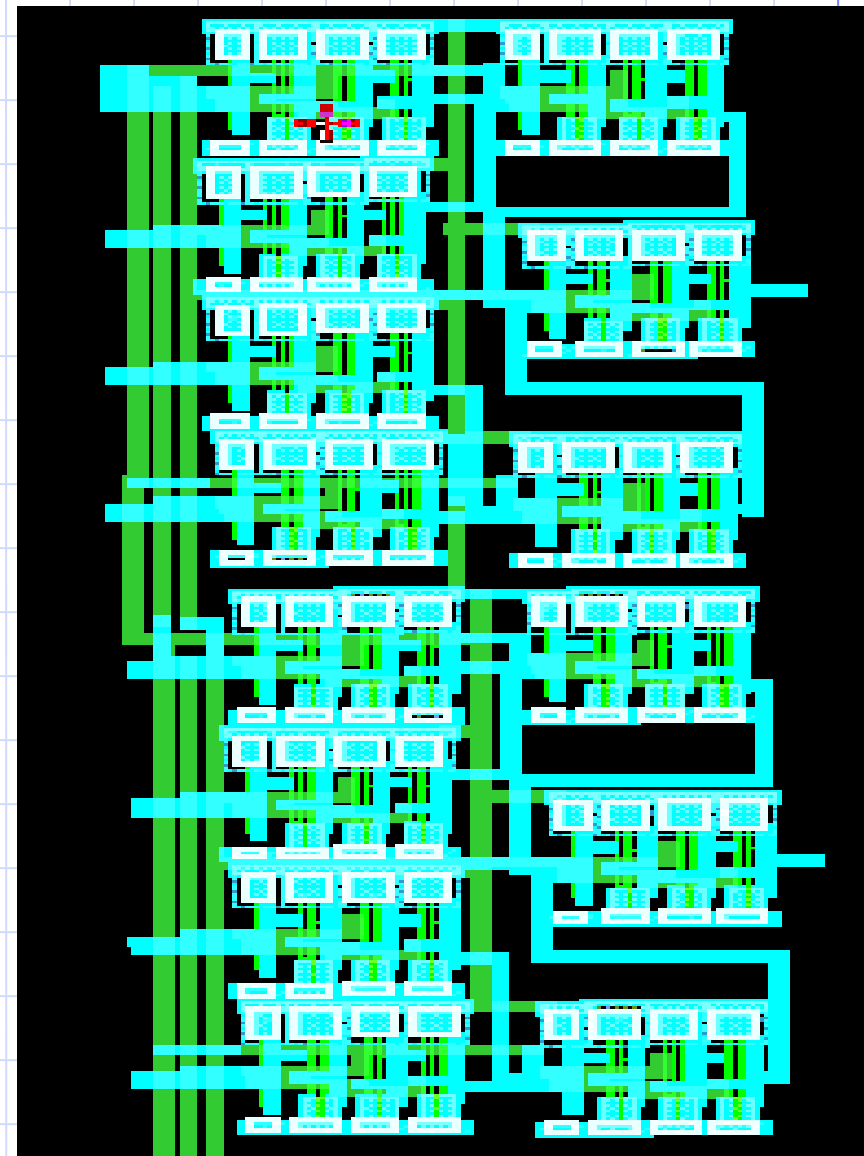
8-bit 8 to 1 Mux



Layout of 1-bit Mux81

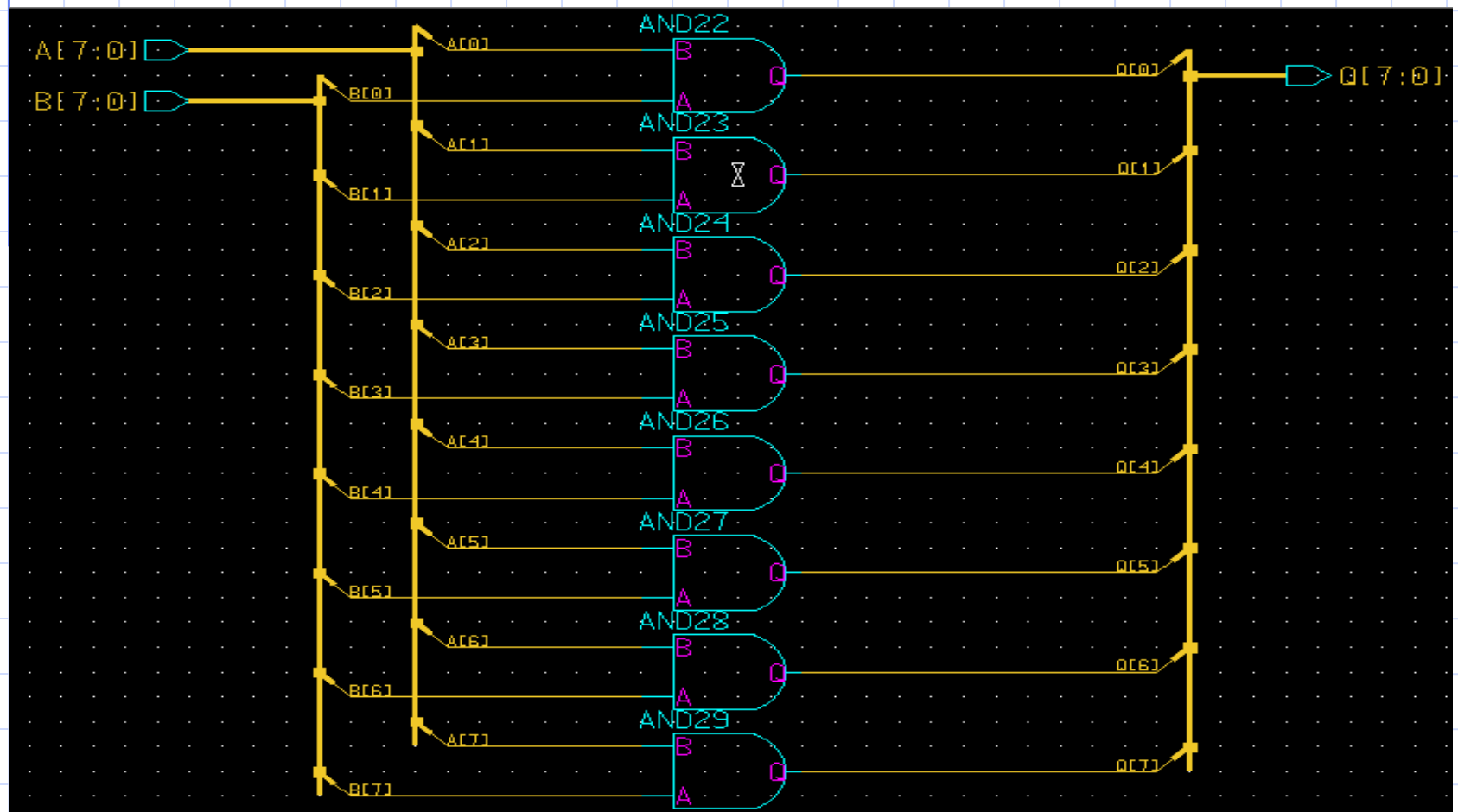
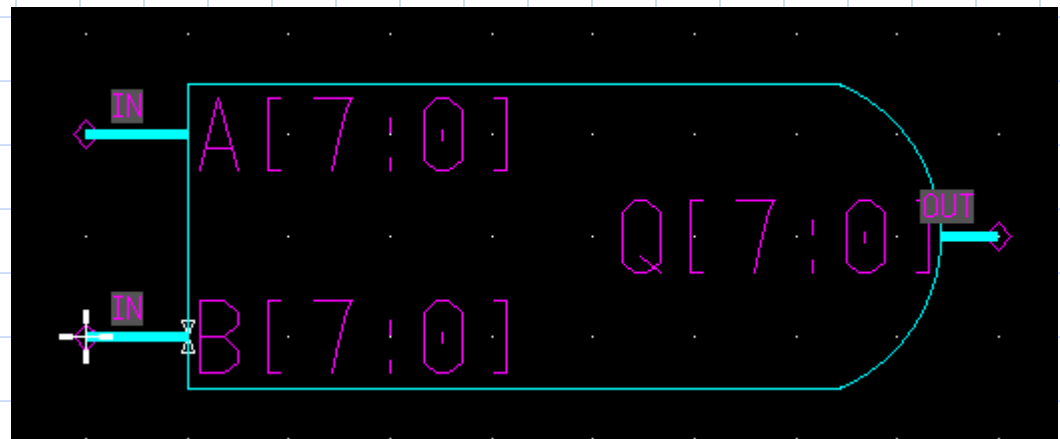


2-bit Mux81



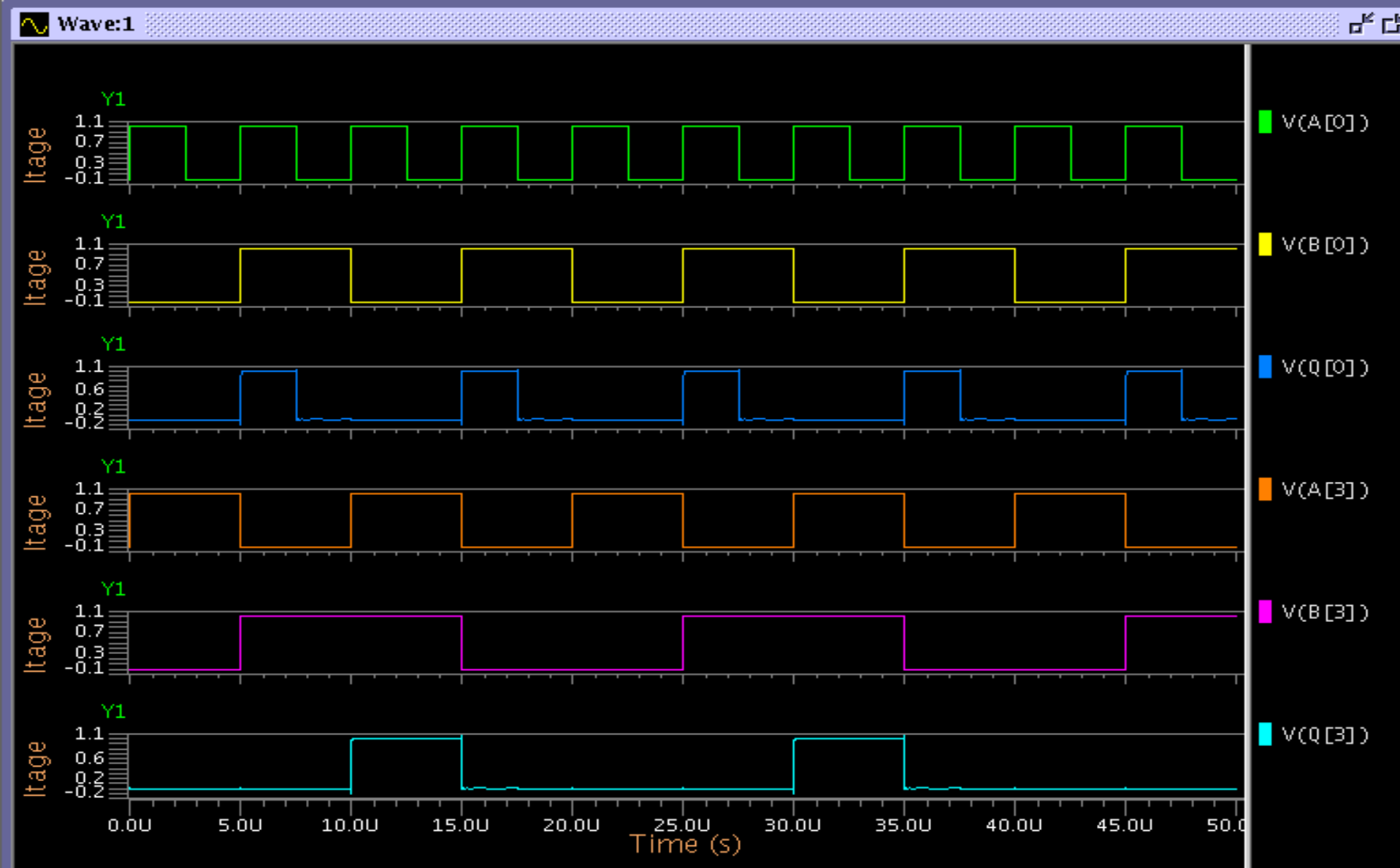
Logic Function

◆ 8-bit AND



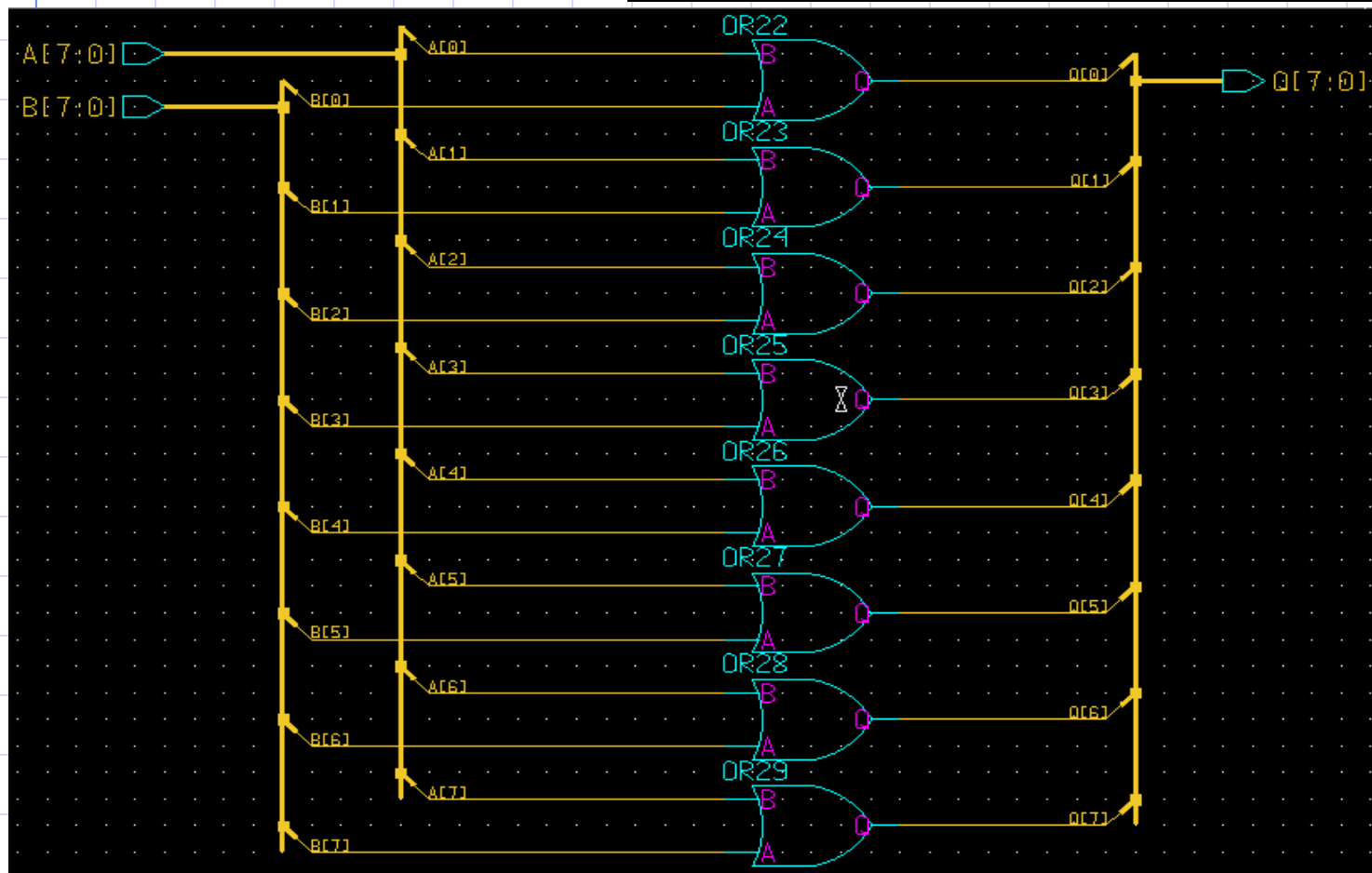
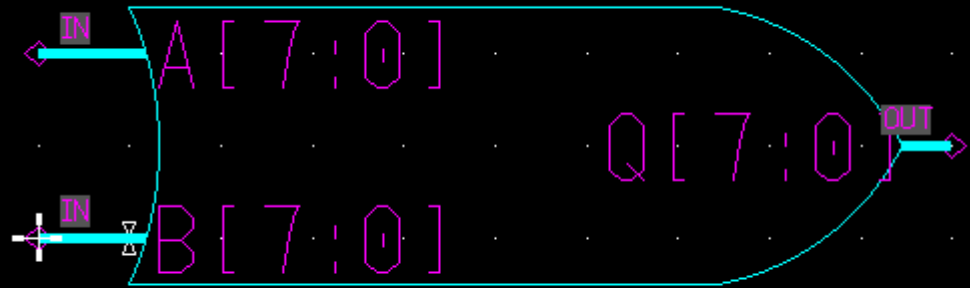
Simulation for 8bit input AND

Input A[7:0]=01010101, B[7:0]=10101010, Q[7:0]=00000000



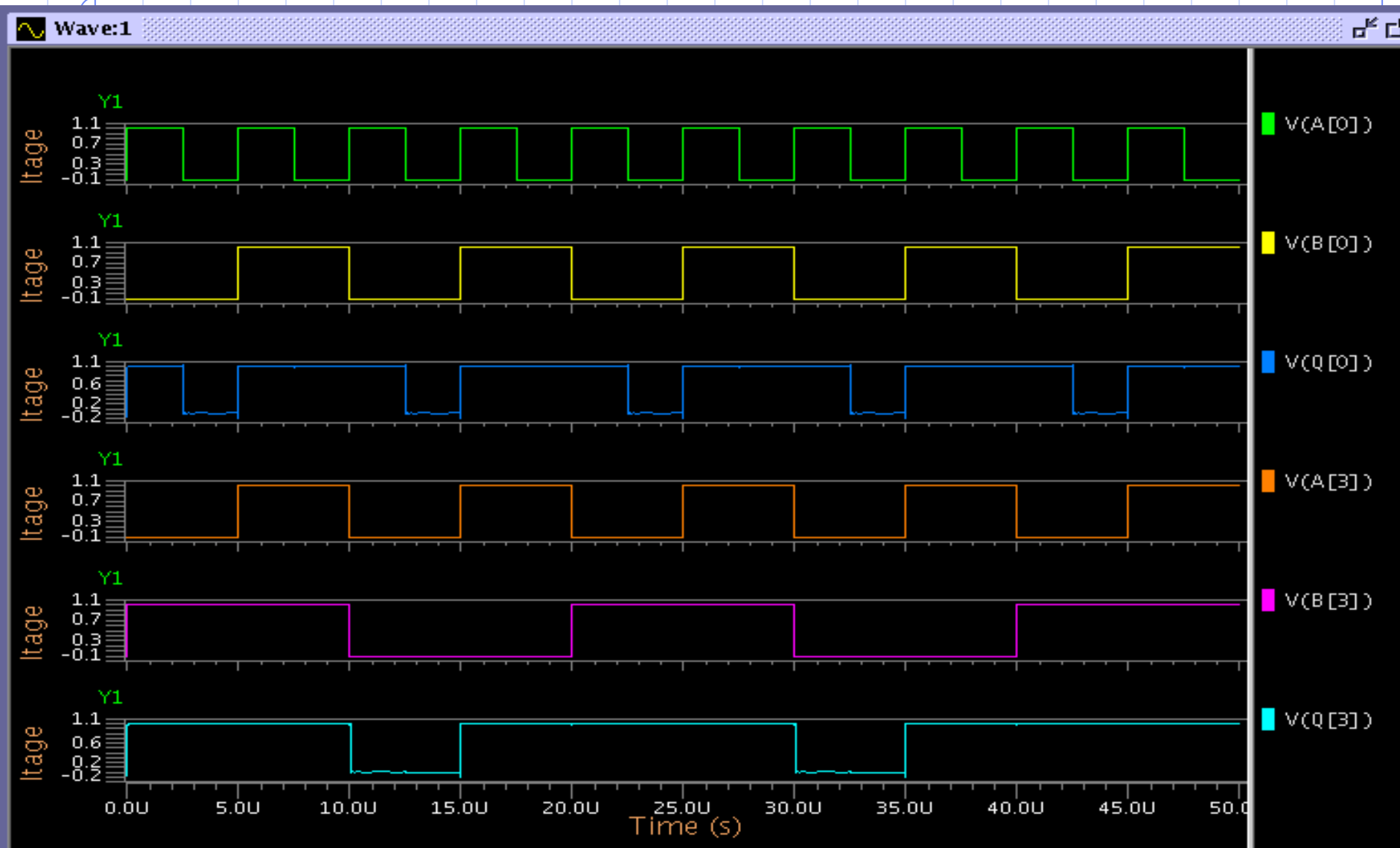
Logic Function

◆ 8bit OR

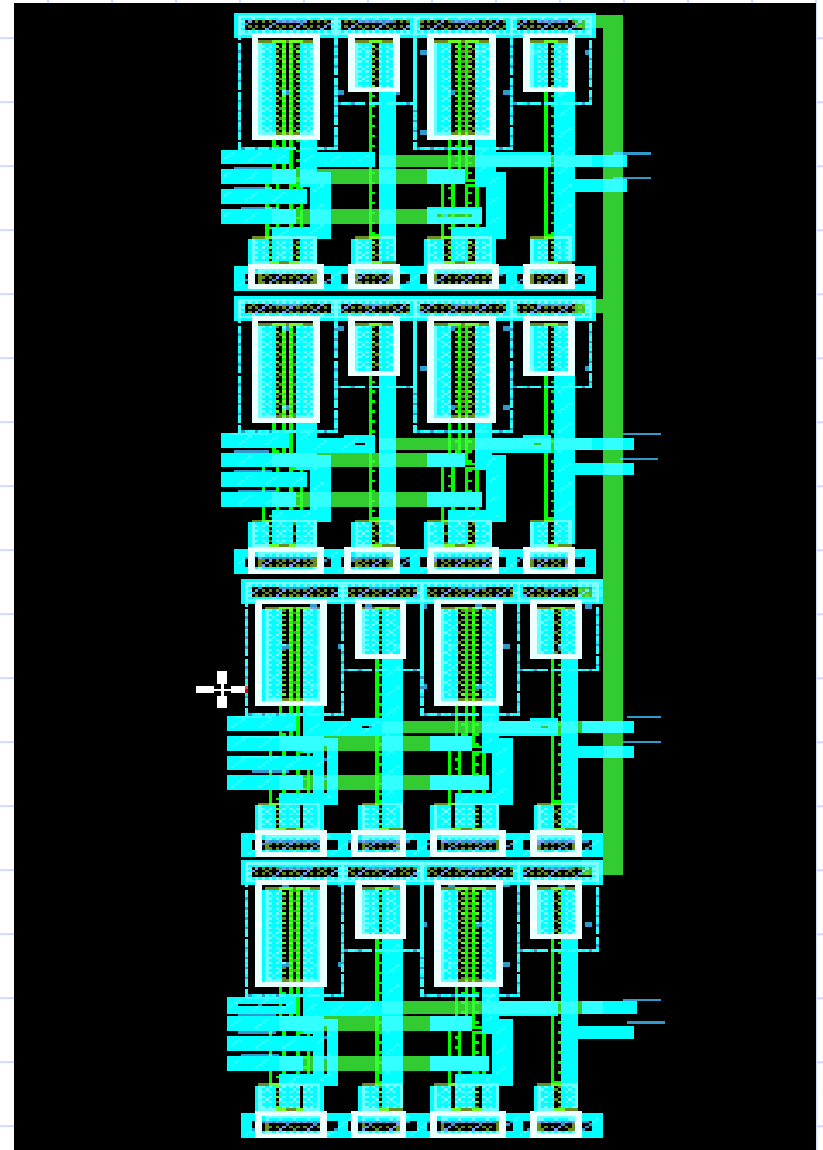
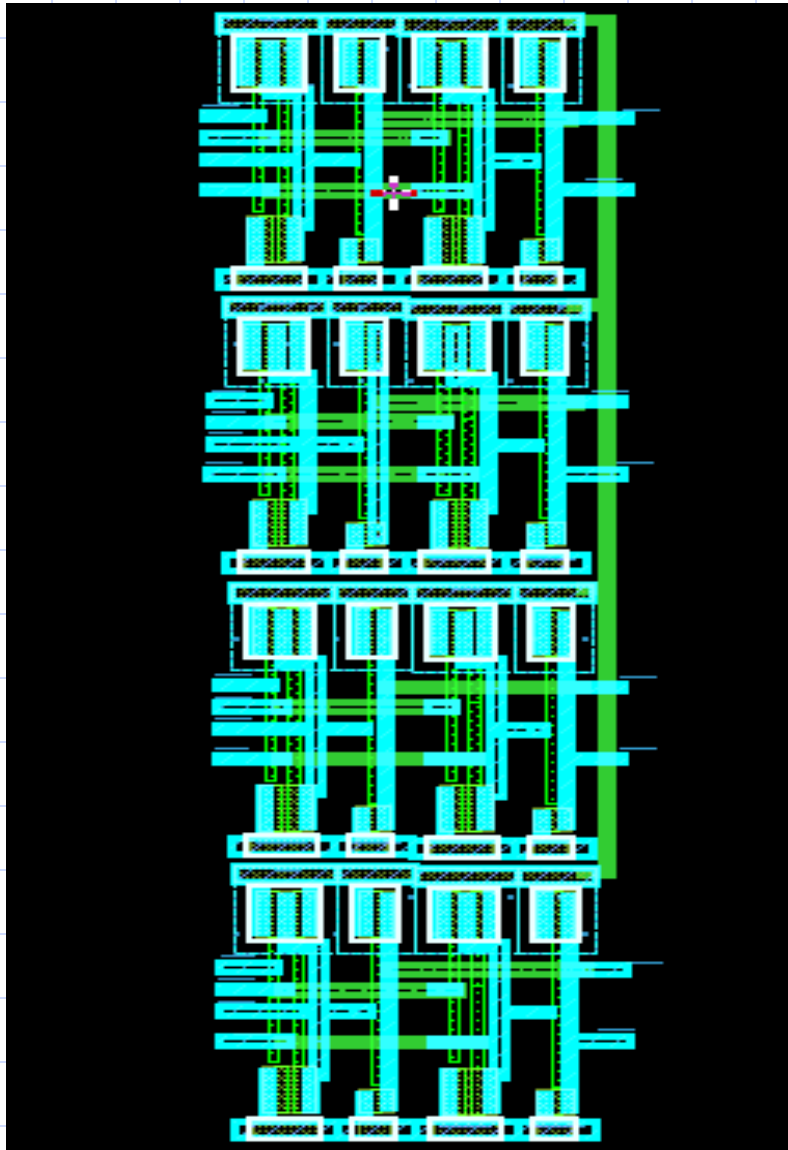


Simulation for 8bit input OR

Input A[7:0]=01010101, B[7:0]=10101010, Q[7:0]=11111111

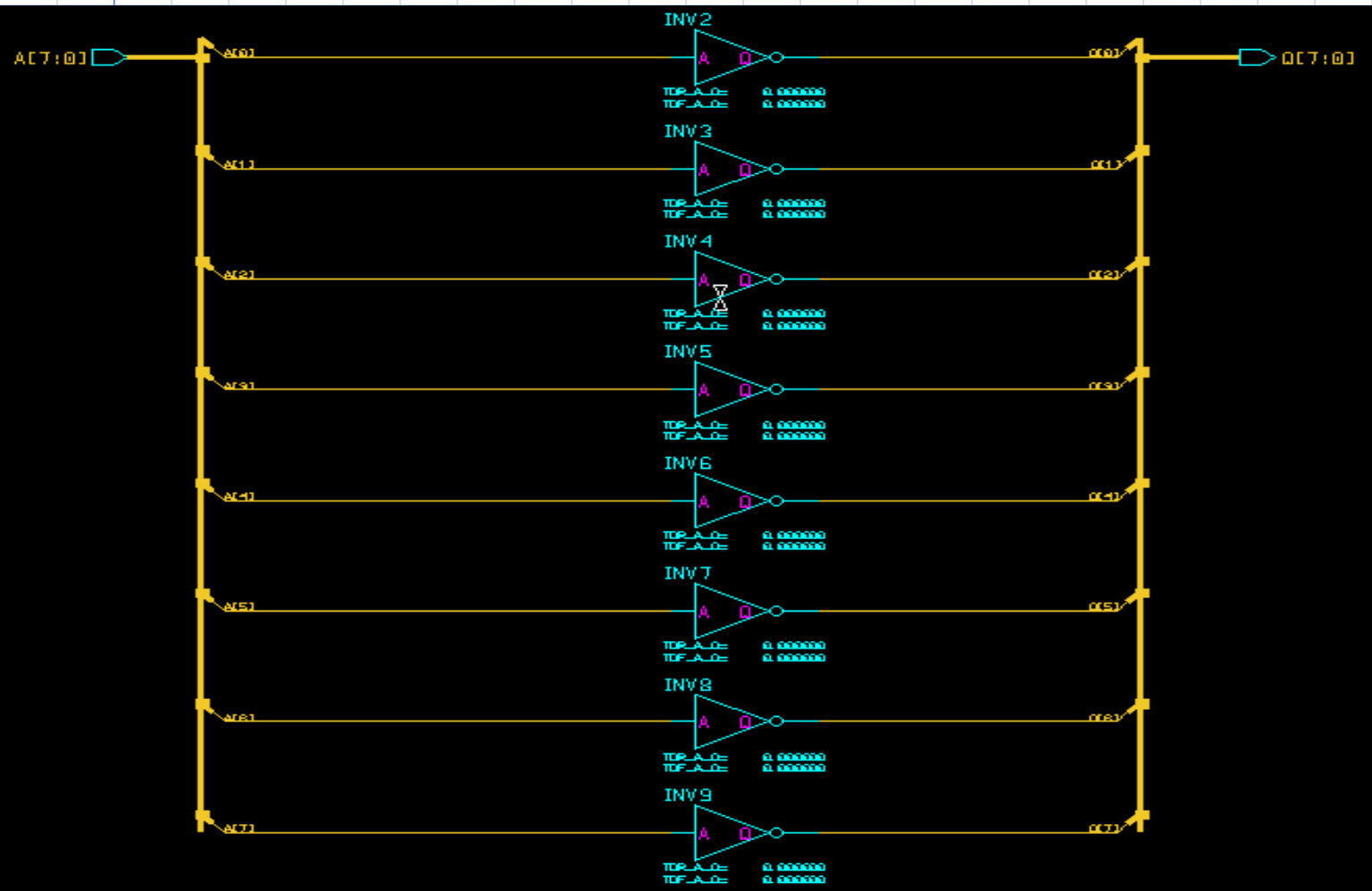
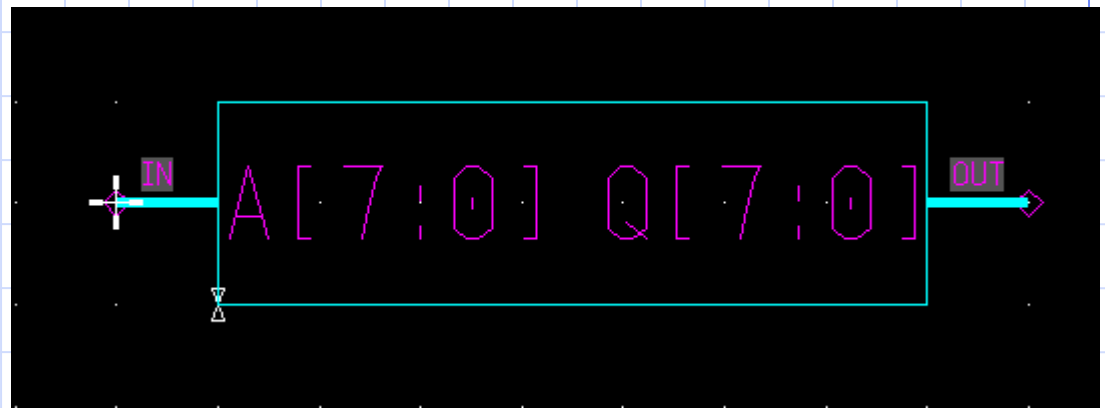


Simulation for 8bit input AND & OR



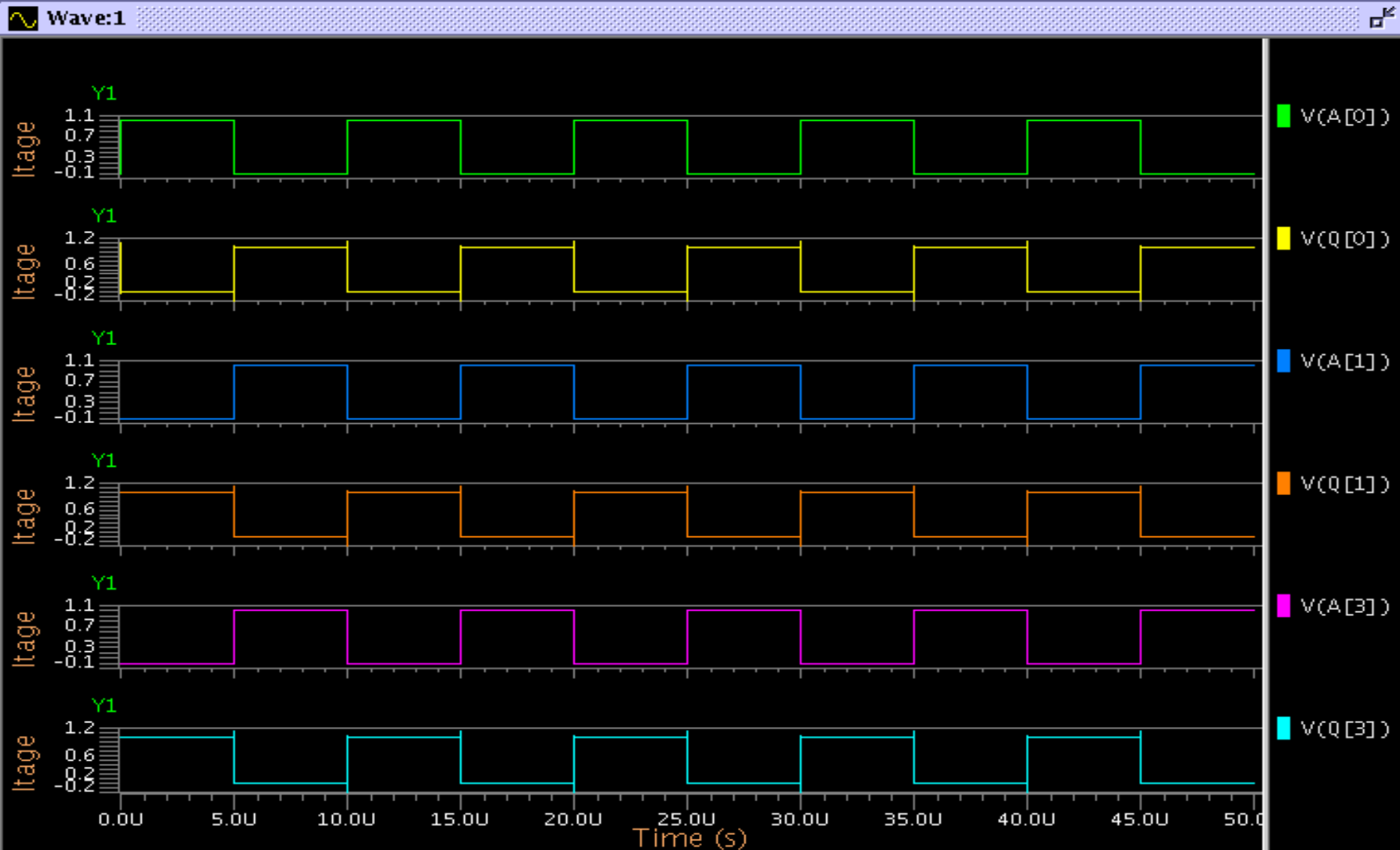
Logic Function

◆ 8bit Invertor



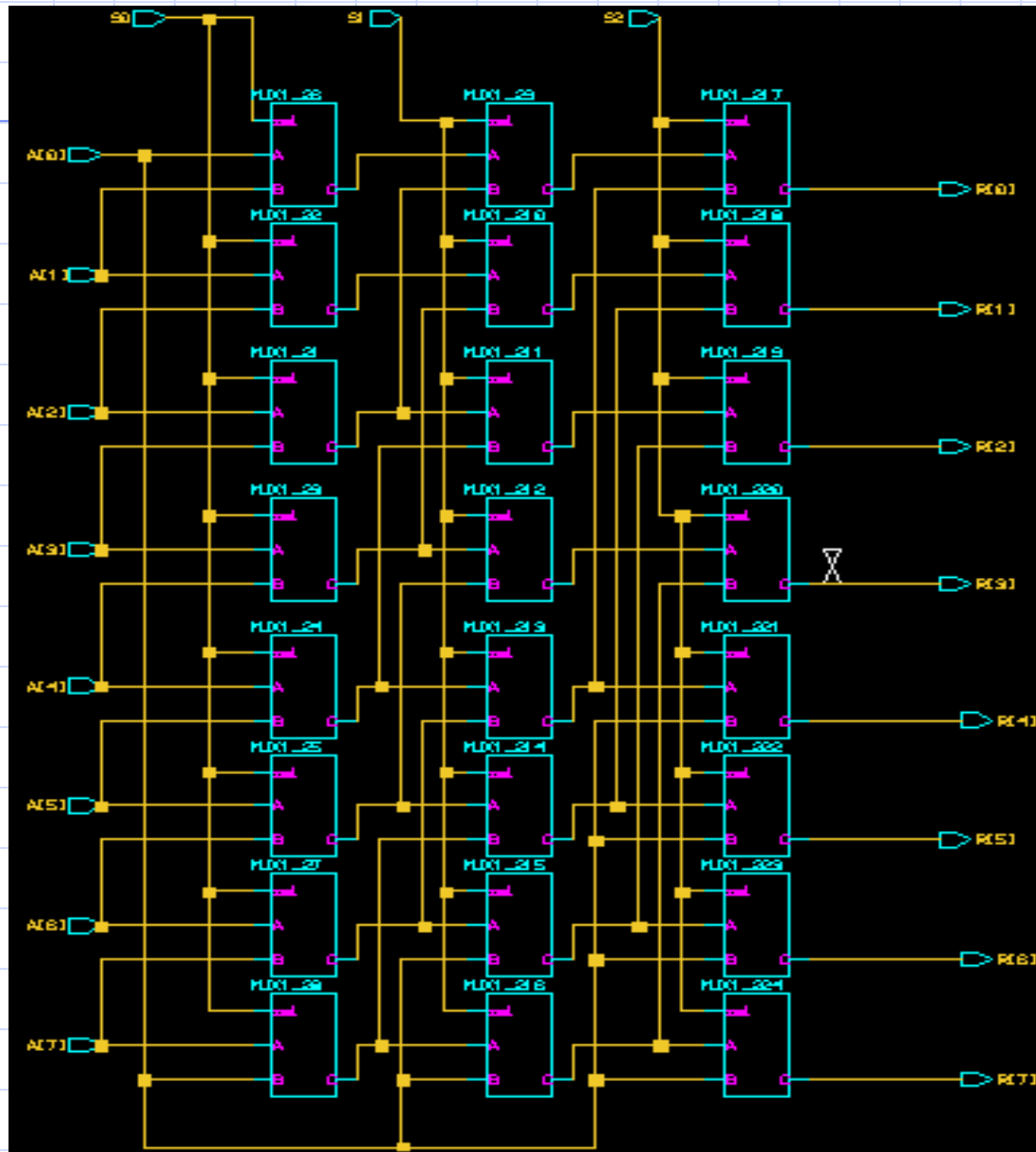
Simulation for 8bit input OR

Input A[7:0]=01010101, Q[7:0]=10101010



Signed-Shift Right

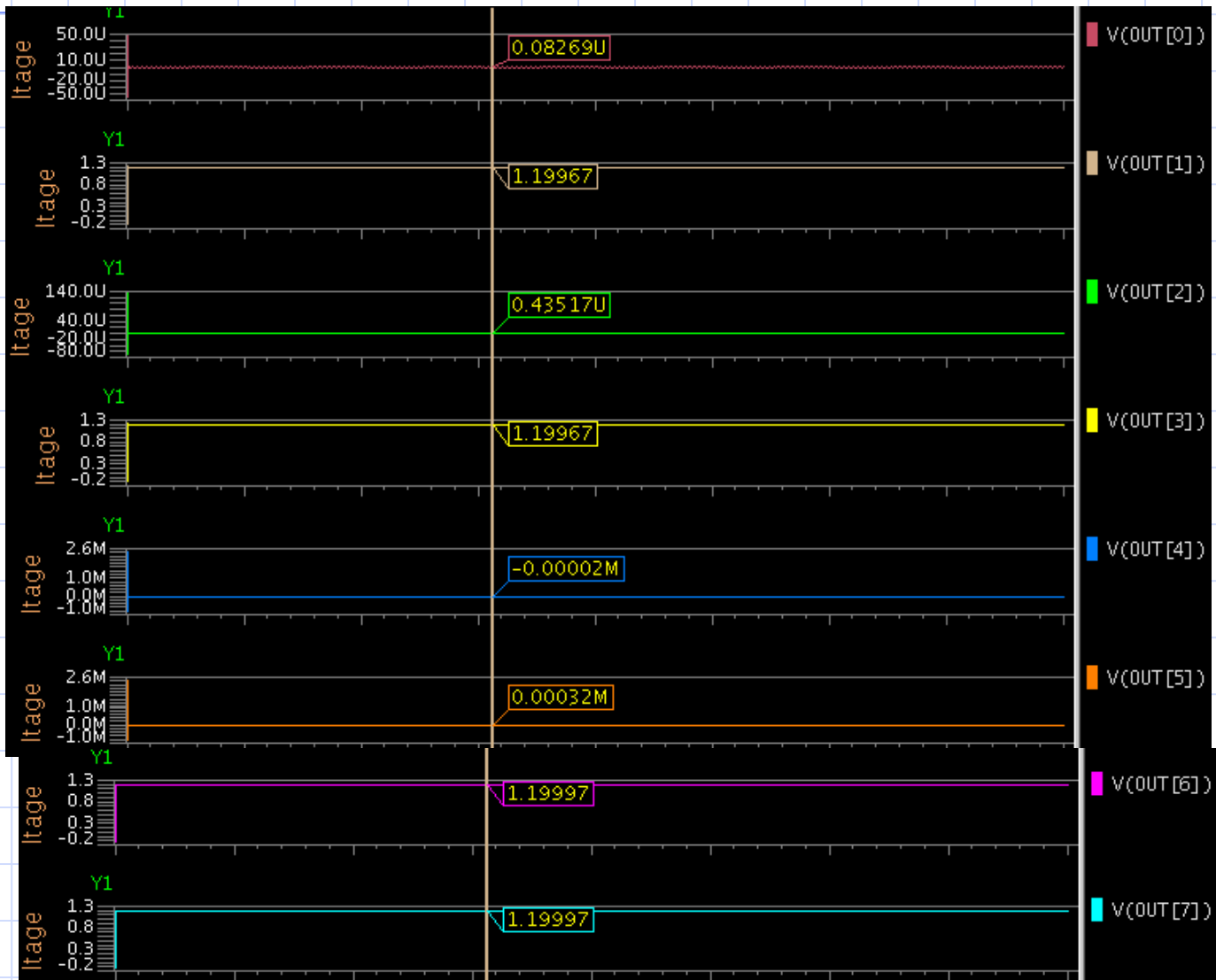
Schematic



Simulation for Signed-shift right

$S_0S_1S_2=010$, connect A0 to extra bit, shift $A7\sim A0=00101011$ right for 2

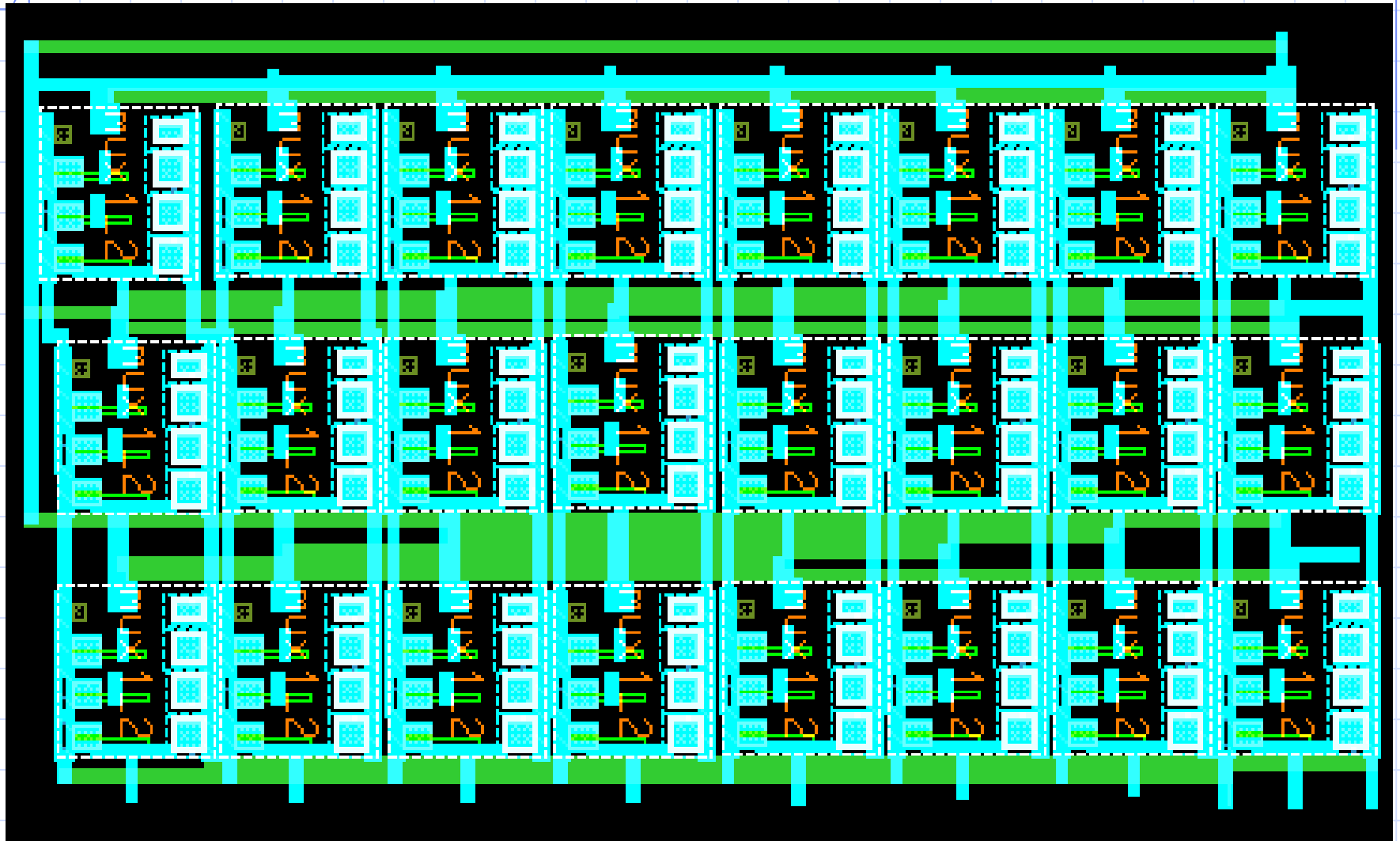
bits, the output $R7\sim R0=11001010$



Signed-shift right

Layout

consist of 24 2-input MUX, using Metal 1 and Metal 2 interconnects



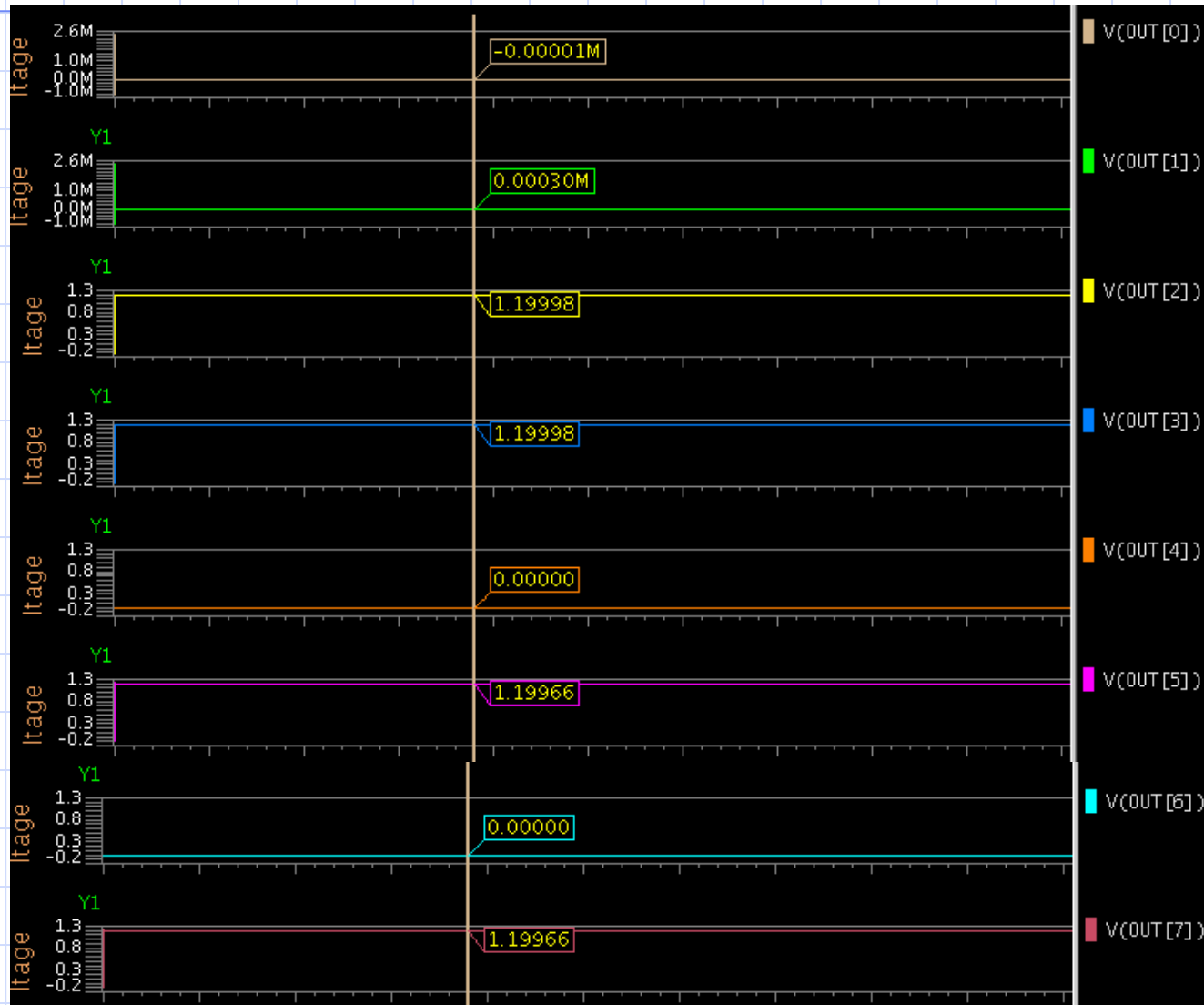
100



Simulation for Signed-shift left

$S_0S_1S_2=010$, connect A7 to extra bit, shift $A7\sim A0=00101011$ left for 2

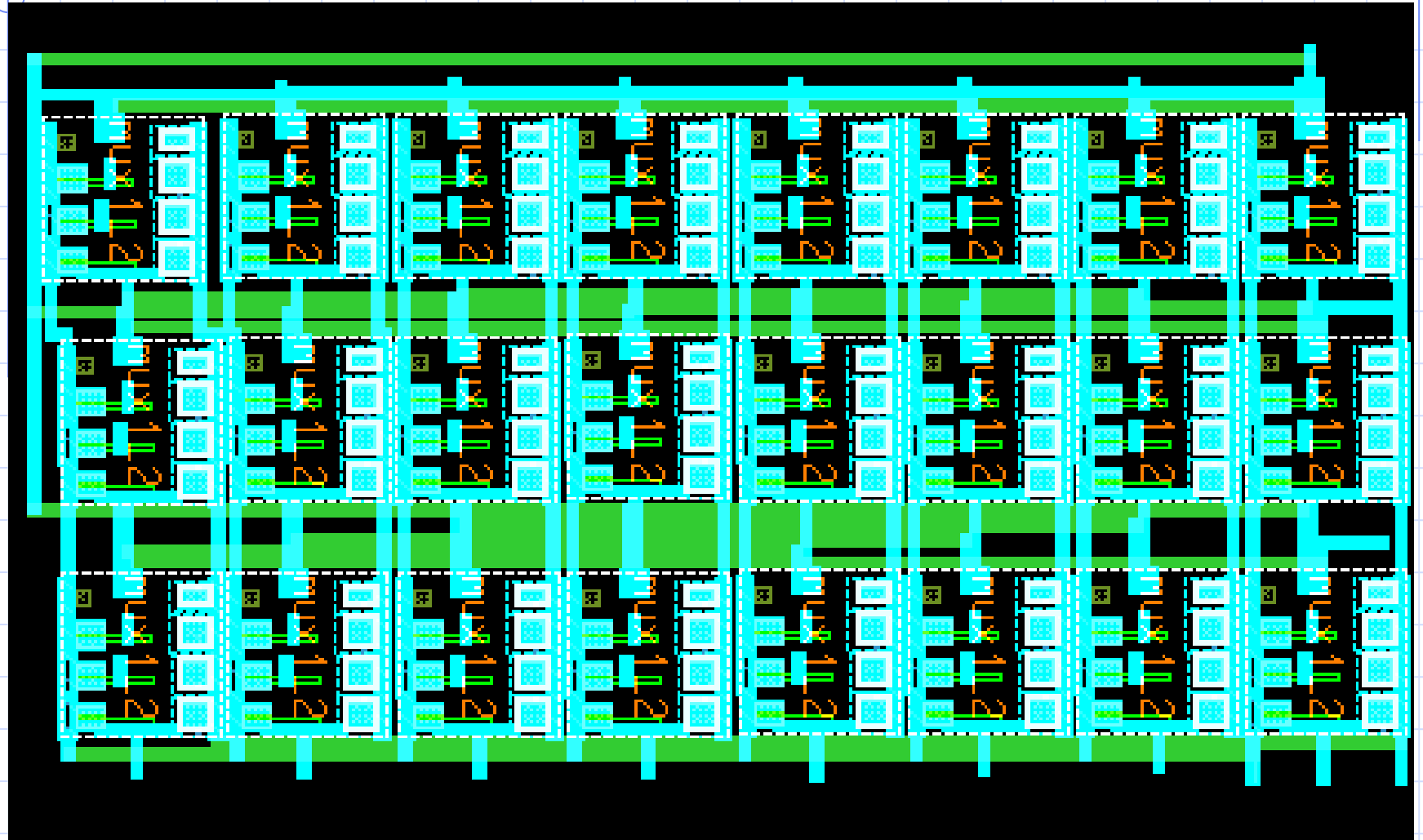
bits, the output $R7\sim R0=10101100$



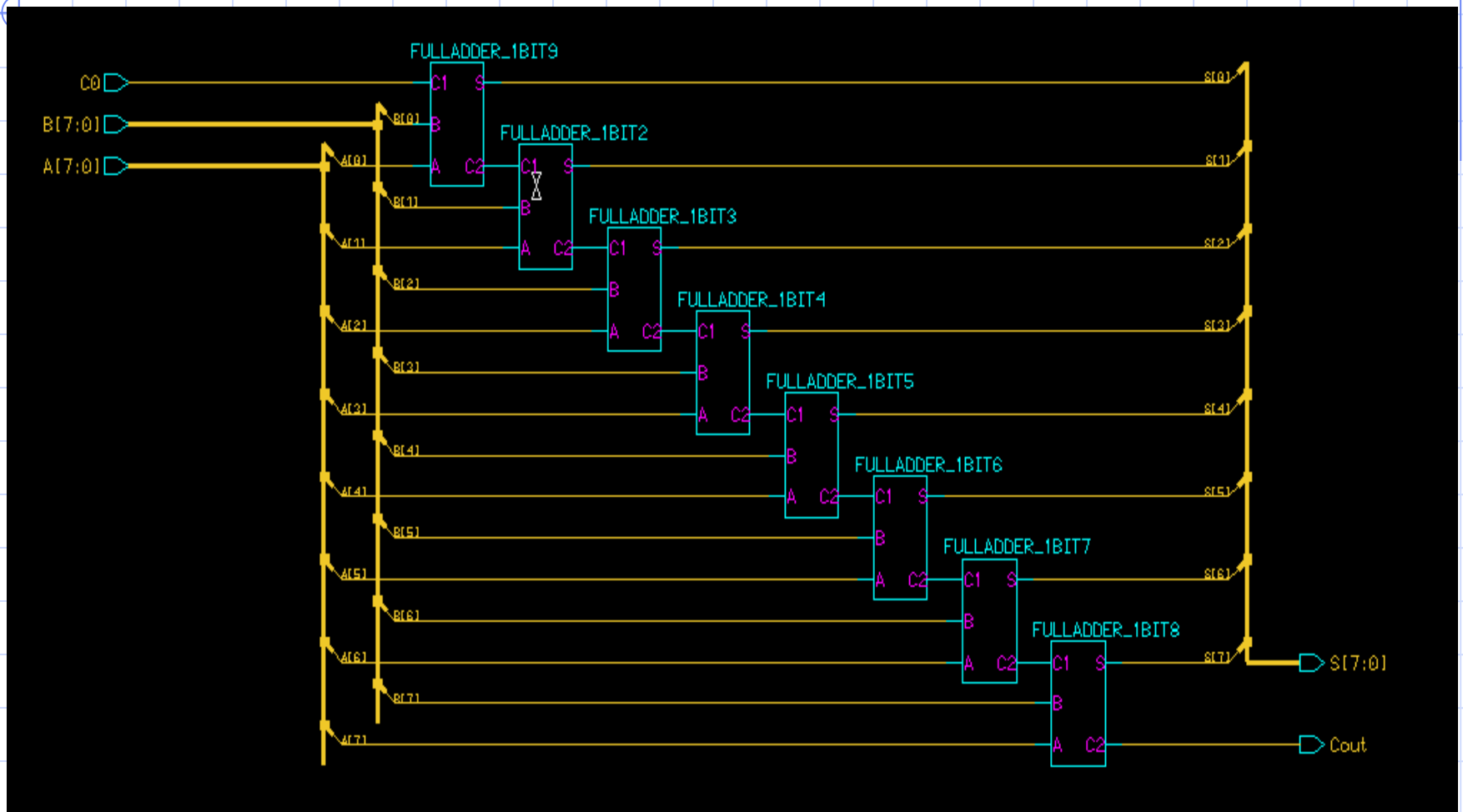
Signed-shift left

Layout

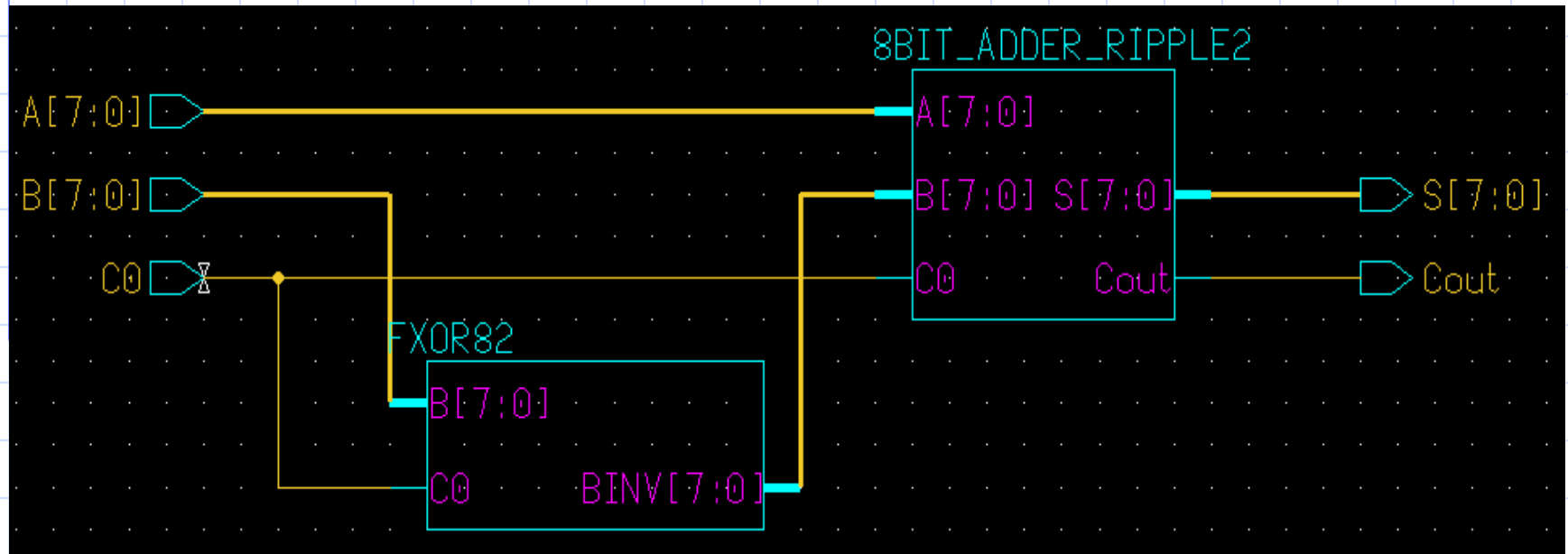
consist of 24 2-input MUX, using Metal 1 and Metal 2 interconnects



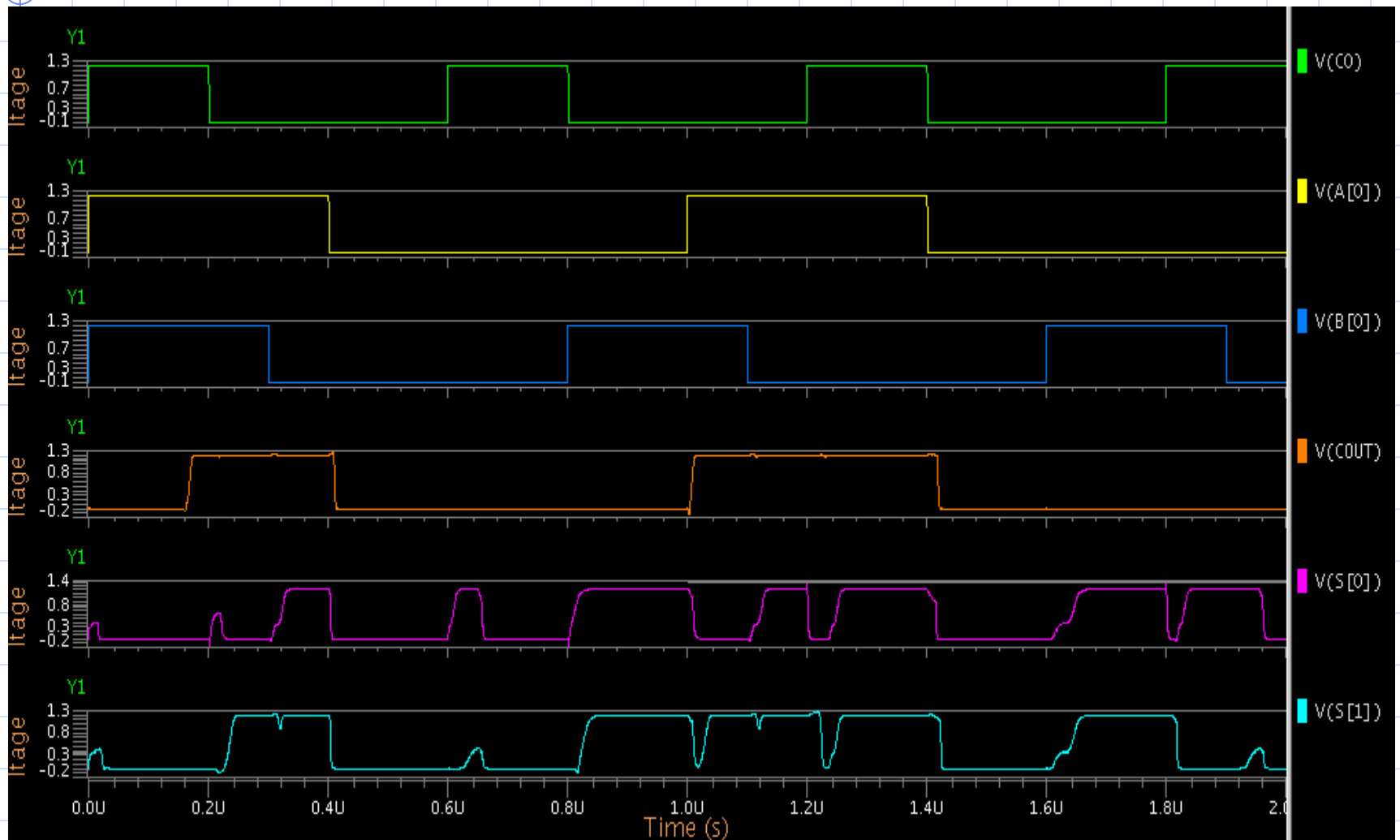
8-bit ripple Adder schematic



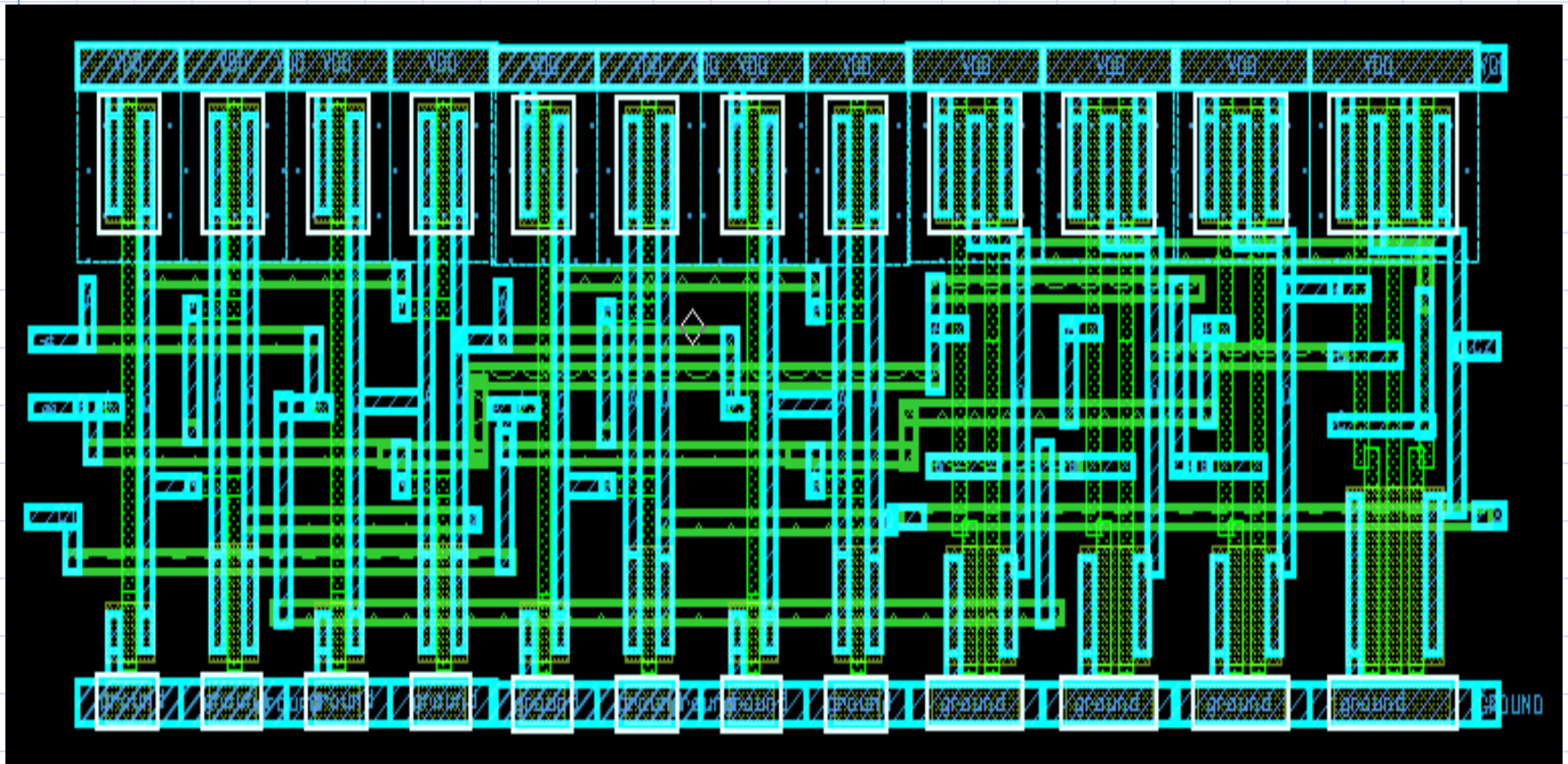
8-bit ripple Add-sub schematic



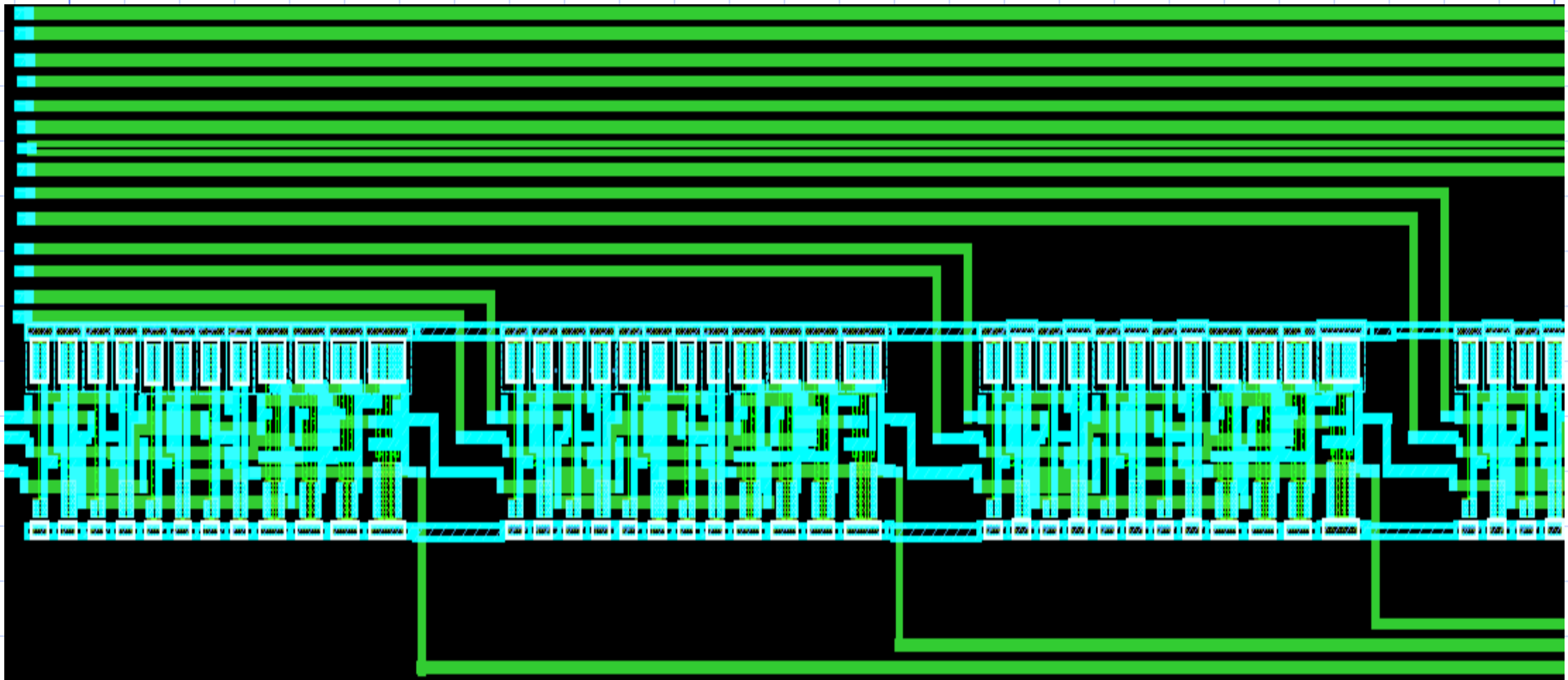
Simulation



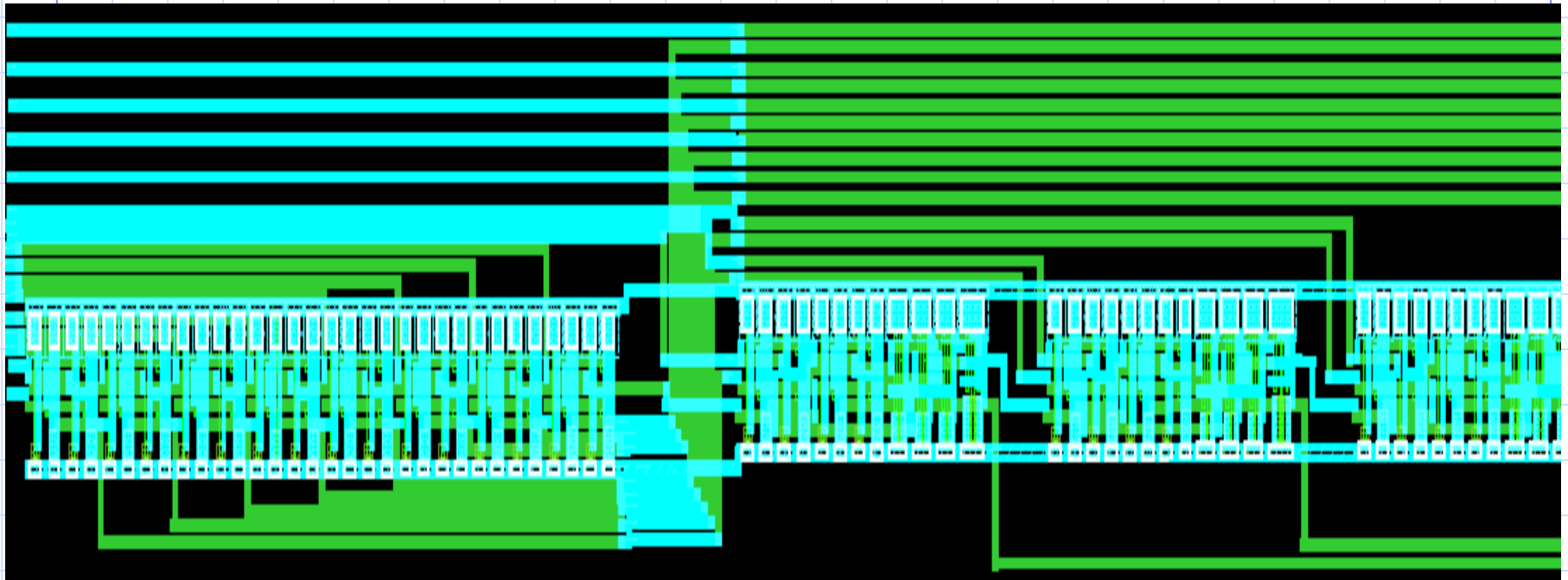
1-bit full adder layout



8-bit ripple adder layout (part)

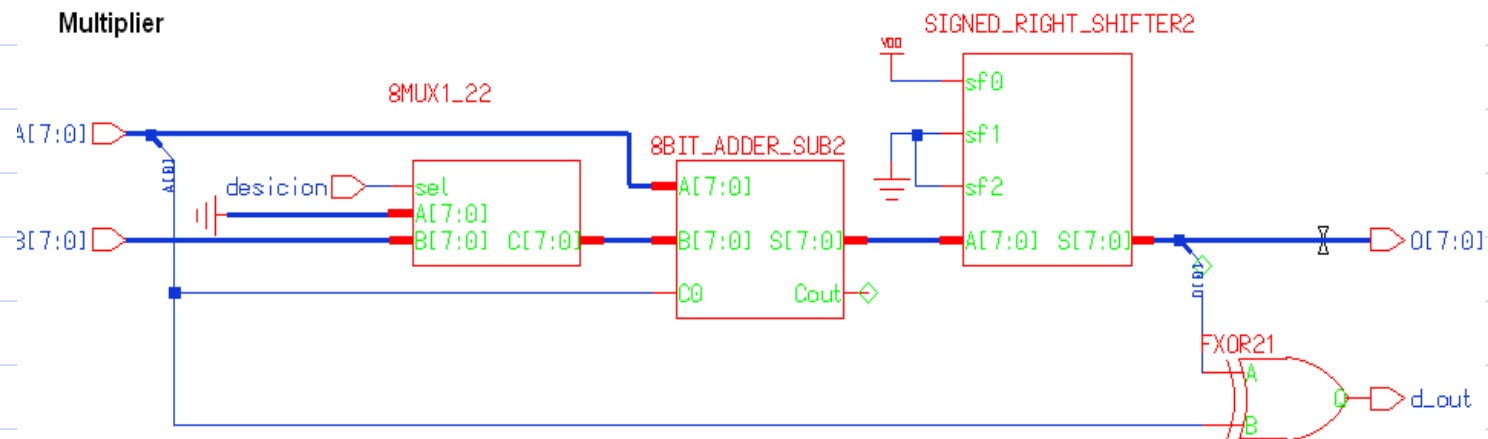
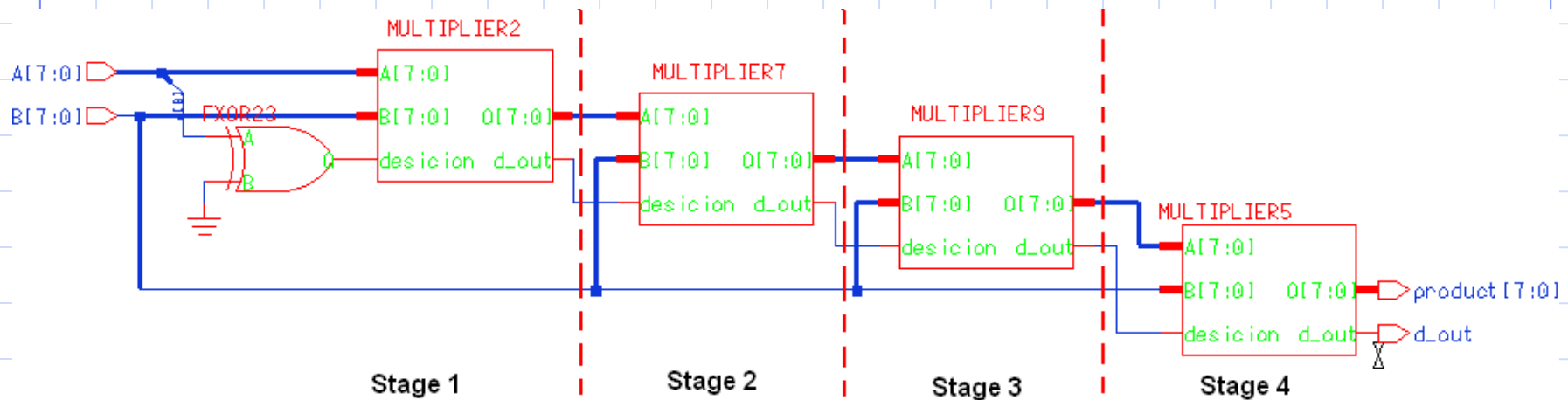


8-bit add-sub layout (part)



4-bit Booth Multiplier

Schematic



4-bit Booth Multiplier

◆ Simulation

$$B * A = 2 * 3$$

Multiplier A ---0000 0011

Multiplicand B ---0010 0000

Stage1: 0000 001**10** subtract B, shift
- 0010 0000

Stage2: 1111 000**11** nothing, shift

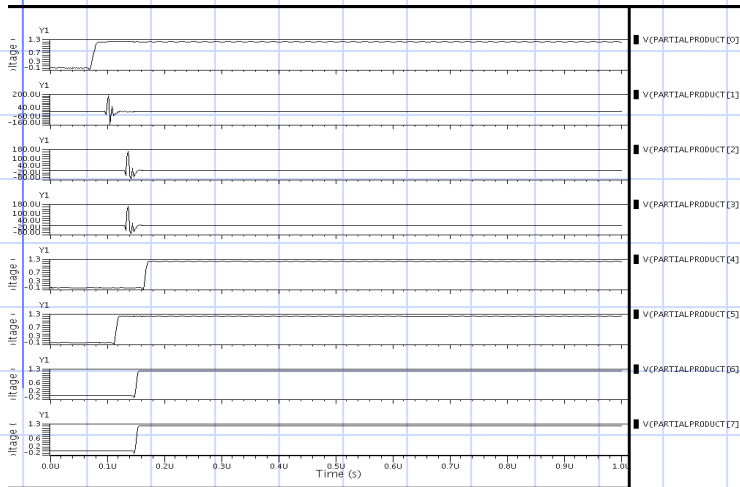
Stage3: 1111 100**01** add B, shift
+ 0010 0000

Stage4: 0000 110**00** nothing, shift

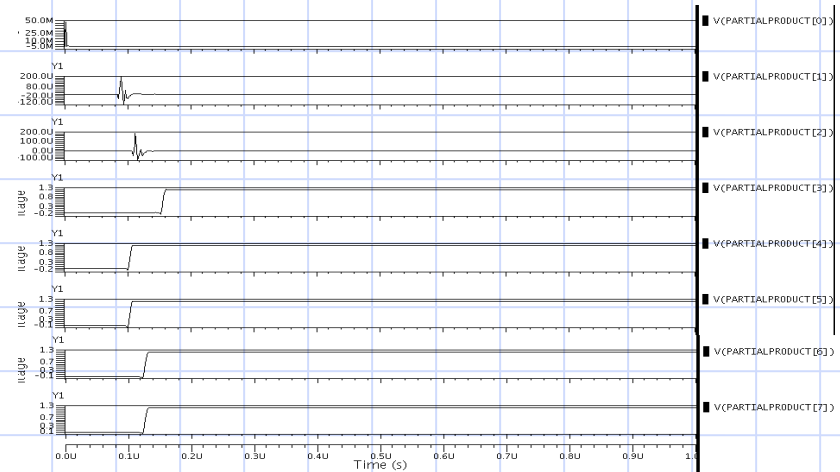
Product: 0000 0110-----done, $6 = 2 * 3$

4-bit Booth Multiplier

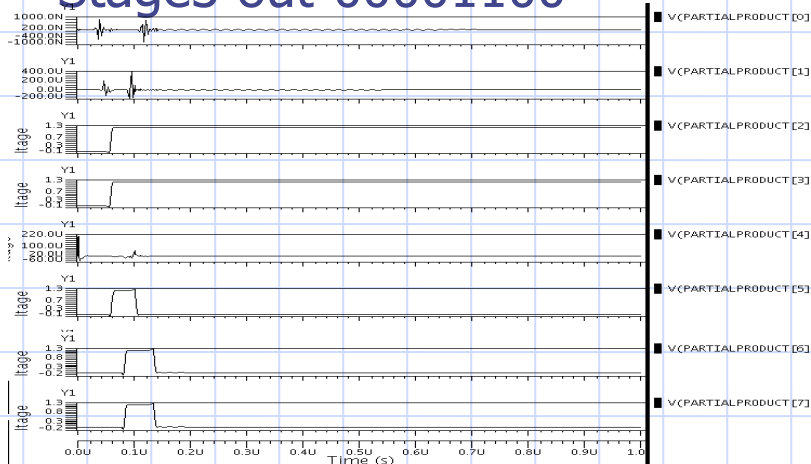
Stage1 out-11110001



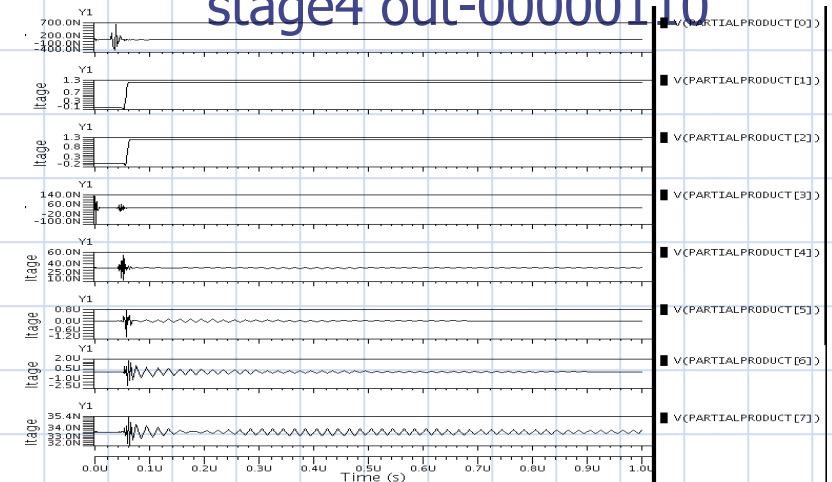
stage2 out-11111000



Stage3 out-00001100

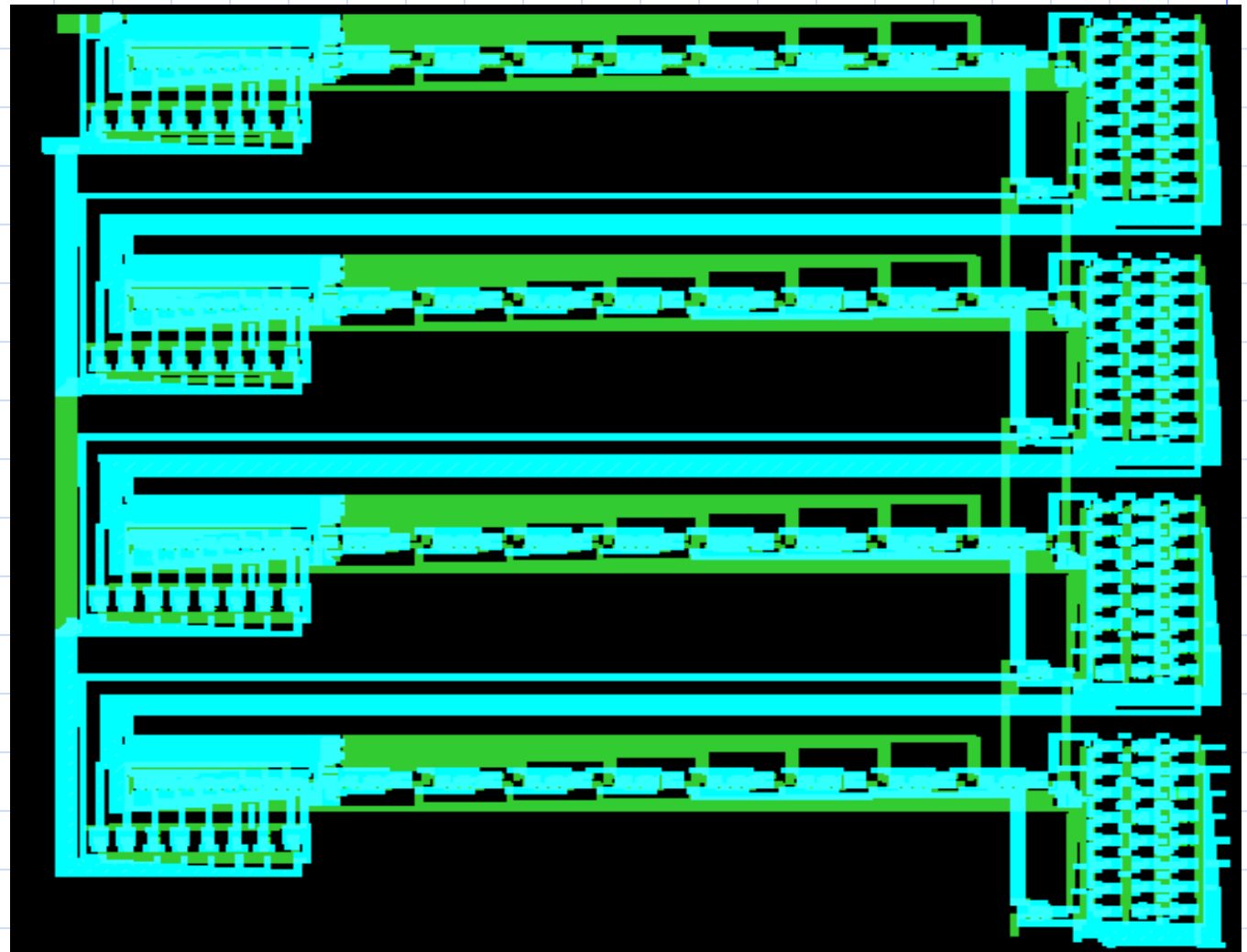


stage4 out-00000110



4-bit Booth Multiplier

◆ Layout



Summary

- ◆ We successfully design 8-bit ALU, supporting 4-bit multiplication.
- ◆ The design has a right simulation result and passes the DRC and LVS verification
- ◆ Final chip:
 - 4464 transistors
 - $S=2\text{mm} \times 2\text{mm}$